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Multi-timescale PTM architecture in human RNA polymerase II

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Highlights

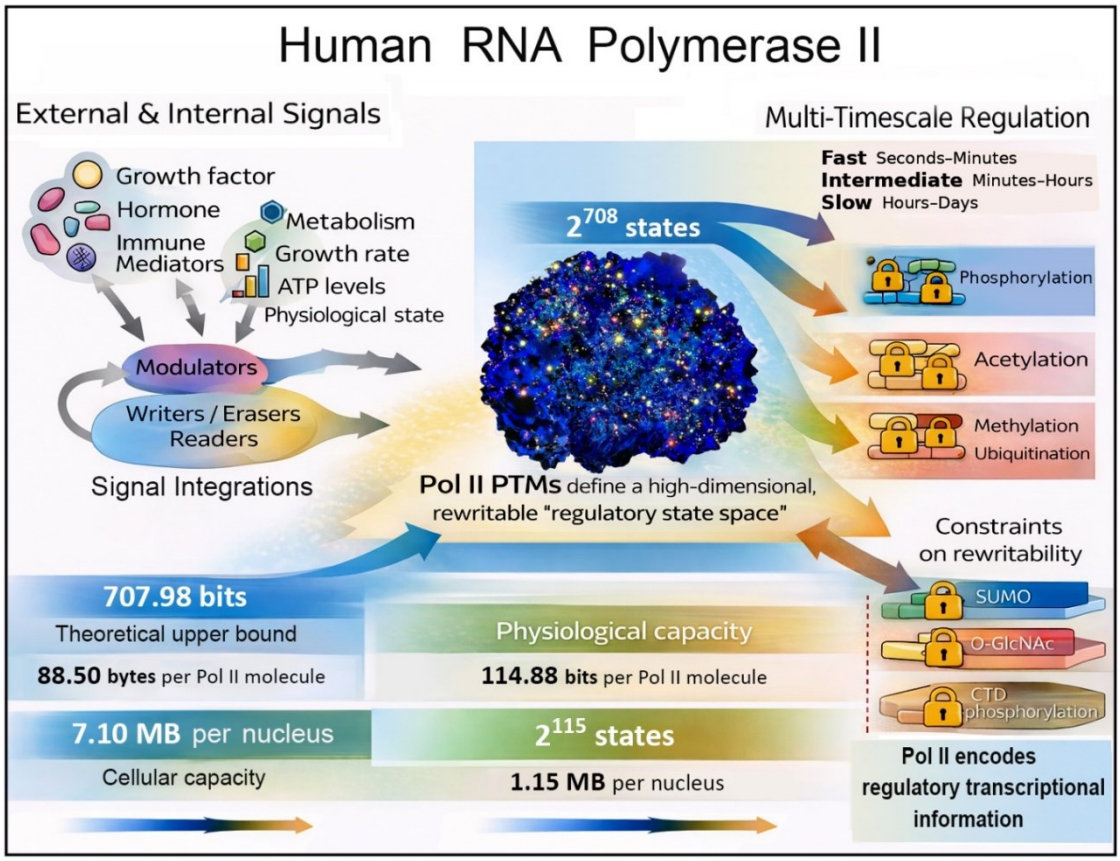
A systems-level framework quantifies regulatory information in Pol II PTMs

Known PTM modification sites provide 707.98 bits per Pol II as a regulatory upper bound

Physiological kinetic constraints reduce accessible regulatory capacity to 114.88 bits

Distinct PTM modification classes define fast, intermediate, and slow regulatory layers

Graphical Abstract



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Summary

Human RNA Polymerase II (Pol II) is characterized by a dense layer of over 775 post-translational modifications (PTMs) that form a dynamic, rewritable regulatory architecture integrating large numbers of cellular signals to coordinate transcription initiation, elongation, termination, and co-transcriptional RNA processing. While genomic information has been extensively catalogued, the potential information capacity associated with Pol II PTM patterns has remained largely unquantified. Here, we analyze the PTM sites across the human Pol II complex (Rpb1–Rpb12) and estimate the state-space information capacity using Shannon entropy theory. We first provide a theoretical upper bound of ∼707.98 bits per Pol II molecule (∼88.50 bytes) corresponding to ∼5.68×10^7 bits per nucleus (∼7.10 MB), assuming ∼80,200 Pol II molecules per cell. We distinguish this maximal capacity from a conservative, kinetically addressable estimate of ∼114.88 bits per molecule (∼1.15 MB per nucleus), reflecting physiological kinetic constraints and site coupling that restrict the simultaneously addressable PTM state space in vivo. Finally, we show that major PTM classes (phosphorylation, proline isomerization, O-GlcNAcylation and ubiquitination) operate over distinct lifetimes, from seconds to minutes and hours-scale processes, supporting a multi-timescale biochemical architecture of this enzyme. Together, these results provide a quantitative information framework that distinguishes maximal PTM state-space capacity from kinetically addressable physiological regulatory capacity, supporting a view of Pol II PTM patterning as a high-dimensional, dynamically reconfigurable, multi-timescale regulatory information layer.

Keywords: Systems Biology; RNA polymerase II; information capacity; CTD code; Shannon entropy; post-translational modifications; kinetic layers.

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Introduction

RNA polymerase II (Pol II) is the central enzyme that converts genomic information into RNA output in human cells. This fundamental cellular process allows DNA to be read and RNA transcripts to be synthesized, including all protein-coding mRNAs and multiple classes of non-coding RNAs (e.g., snRNA, miRNA, and snoRNA), in a manner closely adapted to cellular conditions[1].

To maintain an accurate transcription, Pol II operates within a highly complex regulatory network that helps determine which genomic regions are transcribed, and to what extent, throughout the cellular life, ensuring that only the genomic information necessary for optimal physiological function is transcribed at any given time in the cell cycle[2].

The activity of this sophisticated enzyme requires a continuous supply of chemical energy from nucleotide hydrolysis, which is converted into mechanical work generating forces on the order of tens of piconewtons during transcription elongation, comparable to or exceeding those reported for many ATP-dependent molecular motors[3]. This force pulls the DNA template through the enzyme, allowing sustained elongation rates spanning from sub-kilobase to several kilobases per minute in live-cell estimates[4]. Despite this speed, Pol II maintains high fidelity through proofreading and backtracking mechanisms, yielding low transcriptional error rates[5].

To synthesize the appropriate RNAs at each moment, Pol II acts as one of the most complex regulatory hubs in the cell, integrating a wide range of extracellular and intracellular signals. External stimuli, such as hormones[6], growth factors[7], heat stress[8], developmental cues[9], hypoxia[10], oxidative stress[11], and inflammation[12], modify the transcriptional outcome. Pol II also responds to intracellular variables including ATP availability[13], growth rate[14], metabolic state and nutrient availability[15,16], endocytic signaling[17], endoplasmic reticulum stress responses[18], mitochondrial activity[19], cytoskeletal reorganization[20], cell-cycle state[21], chromatin organization[22], circadian programs[23], and DNA damage responses[24].

Mechanistically, Pol II operates within a large and complex dynamic molecular network that primarily involves the general transcription factors, the Mediator complex, chromatin remodelers, RNA-processing machinery, elongation/pausing regulators, and termination processes[25]. Throughout the transcription cycle, Pol II engages numerous enzymatic activities, including kinases, phosphatases, helicases, translocases, methyltransferases, acetyltransferases, exonucleases, ubiquitin ligases, and cyclin-dependent kinases, that continuously tune transcriptional dynamics in response to cellular context[26–30]. As a result, individual human cells exhibit rapid and continuous transcriptional adaptation, producing thousands of nascent transcripts at any given time[31].

A key reason Pol II can integrate so many inputs is its extensive and reversible layer of post-translational modifications (PTMs), in which chemical groups such as phosphate, methyl, or acetyl moieties are covalently attached to specific amino acid residues. These modifications can rapidly reconfigure interaction surfaces, alter the binding avidity of regulatory proteins, and modulate transitions between transcriptional steps[32]. In this sense, Pol II can encode regulatory information through combinatorial patterns of PTMs that are written, erased, and interpreted by dedicated enzymes and reader proteins, enabling context-dependent control of transcriptional output[33–35]. As a consequence, these covalent PTM marks provide a rapid and reversible mechanism for integrating environmental and intracellular signals into the structure and functionality of Pol II[36], establishing an information-rich molecular landscape that orchestrates all phases of the transcription cycle[32].

Despite significant progress in cataloging covalent modifications in Pol II[37–41], a central question remains unanswered: what is the potential information capacity of these biochemical covalent marks? While the

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genomic information content has been studied, for example, in ENCODE Project Consortium[42], a comparable quantification for Pol II is lacking. Without such quantification, the magnitude of this regulatory layer remains conceptually imprecise, despite strong links between PTM dysregulation and disease-related phenotypes[43].

Here we address this gap using Shannon information theory applied to residue-level PTM state spaces. We compiled the human Pol II PTM sites from the experimental scientific literature, publicly available databases and defined explicit discrete-state models for major PTM classes. Under independent and equal-probability assumptions, we estimate an upper-bound theoretical capacity of ~707.98 bits per Pol II molecule, equivalent to ~5.68×10^7 bits per nucleus (~7.10 MB) given ~80,200 Pol II molecules per cell. We further separate this theoretical capacity from a physiological/kinetically accessible *in vivo* bound, estimated at ~114.88 bits per Pol II molecule (~9.21×10^6 bits per nucleus, ~1.15 MB, assuming ~80,200 Pol II), reflecting that only a subset of all covalent marks can be independently established and stably maintained within a given time window.

Furthermore, we show that these covalent biochemical marks span distinct lifetimes revealing a complex multi-timescale architecture. In fact, phosphorylation is widely understood as encoding regulatory responses[44] lasting ~5 seconds to >15 minutes[45,46], proline isomerization[47,48] and O-GlcNAcylation[49–51] form a mesoscale temporal structure spanning from ~1 min to ~30 minutes, and ubiquitination establishes a slower transient process lasting from ~15 min to ~24 hours[52,53].

This information-theoretic perspective positions Pol II not only as an exceptional transcriptional machine but also as a complex information-processing system across multiple timescales. The framework is grounded in curated PTM resources and a broad experimental literature. All numerical results shown here are fully reproducible using the Python Modules provided in Supplementary Information S4, which regenerate all tables, PTM counts, capacity values, physiological estimates, sensitivity scenarios, and discrete-state models.

Results

We have systematically analyzed the canonical residue-level PTM states in human Pol II and calculated their Shannon information content. Specifically, this ~500 kDa enzyme comprises 12 subunits (Rpb1–Rpb12) that form a highly conserved catalytic core, structurally and mechanistically maintained from archaea to humans[54]. The largest subunit, Rpb1, contains the extended C-terminal domain (CTD), which serves as the main regulatory scaffold for Pol II. The CTD undergoes extensive, reversible PTMs enabling it to coordinate virtually all phases of the transcription cycle, including initiation, promoter escape, early and productive elongation, termination, and co-transcriptional RNA processing[44] (see for more details Supplementary Information S1).

I. Combinatorial PTM state space of the mammalian CTD

The C-terminal domain (CTD) of Rpb1 comprises 52 Tyr-starting heptad repeats related to the consensus Tyr1–Ser2–Pro3–Thr4–Ser5–Pro6–Ser7 (YSPTSPS) and a 10-residue C-terminal motif[55]. In the canonical human sequence only 21 repeats match the exact consensus, whereas the remaining 31 are nonconsensus with substitutions primarily at positions 2, 4, 5, and especially 7; notably, eight of them contain Lys (K7) and two contain Arg (R4 and R7) (The canonical UniProt POLR2A sequence, P24928[37]).

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These CTD residues (Table 1) are extensively modified by multiple PTM types, generating an exceptionally large combinatorial regulatory space. The covalent modifications considered include phosphorylation, acetylation, O-GlcNAcylation, mono-, di- and trimethylation, asymmetric dimethylation, symmetric dimethylation, SUMOylation and ubiquitination (retrieved from PTM resources UniProt[37], iPTMnet[38], PhosphoSitePlus[39], GlyGen[40] and dbPTM[41]), and proline isomerization[47,48,56–60], two additional O-GlcNAcylation marks[49], and citrullination[61] (compiled from the literature).

Table 1. CTD structure comprising 52 heptapeptides and a 10-residue C-terminal tail (UniProt P24928 canonical sequence).

n	Start	Heptad	Class	n	Start	Heptad	Class
1	1593	YSPTSPA	Nonconsensus	**28**	1783	YSPTSPS	Consensus
2	1600	YEPRSPGG*	Nonconsensus	**29**	1790	YSPTSPS	Consensus
3	1608	YTPQSPS	Nonconsensus	**30**	1797	YSPTSPS	Consensus
4	1615	YSPTSPS	Consensus	**31**	1804	YSPSSPR	Nonconsensus
5	1622	YSPTSPS	Consensus	**32**	1811	YTPQSPT	Nonconsensus
6	1629	YSPTSPN	Nonconsensus	**33**	1818	YTPSSPS	Nonconsensus
7	1636	YSPTSPS	Consensus	**34**	1825	YSPSSPS	Nonconsensus
8	1643	YSPTSPS	Consensus	**35**	1832	YSPASPK	Nonconsensus
9	1650	YSPTSPS	Consensus	**36**	1839	YTPTSPS	Nonconsensus
10	1657	YSPTSPS	Consensus	**37**	1846	YSPSSPE	Nonconsensus
11	1664	YSPTSPS	Consensus	**38**	1853	YTPTSPK	Nonconsensus
12	1671	YSPTSPS	Consensus	**39**	1860	YSPTSPK	Nonconsensus
13	1678	YSPTSPS	Consensus	**40**	1867	YSPTSPK	Nonconsensus
14	1685	YSPTSPS	Consensus	**41**	1874	YSPTSPT	Nonconsensus
15	1692	YSPTSPS	Consensus	**42**	1881	YSPTTPK	Nonconsensus
16	1699	YSPTSPS	Consensus	**43**	1888	YSPTSPT	Nonconsensus
17	1706	YSPTSPS	Consensus	**44**	1895	YSPTSPV	Nonconsensus
18	1713	YSPTSPS	Consensus	**45**	1902	YTPTSPK	Nonconsensus
19	1720	YSPTSPS	Consensus	**46**	1909	YSPTSPT	Nonconsensus
20	1727	YSPTSPS	Consensus	**47**	1916	YSPTSPK	Nonconsensus
21	1734	YSPTSPS	Consensus	**48**	1923	YSPTSPT	Nonconsensus
22	1741	YSPTSPN	Nonconsensus	**49**	1930	YSPTSPKGST*	Nonconsensus
23	1748	YSPTSPN	Nonconsensus	**50**	1940	YSPTSPG	Nonconsensus
24	1755	YTPTSPS	Nonconsensus	**51**	1947	YSPTSPT	Nonconsensus
25	1762	YSPTSPS	Consensus	**52**	1954	YSLTSPA	Nonconsensus
26	1769	YSPTSPN	Nonconsensus	**Tail**	1961	ISPDDSDEEN	Terminal tail
27	1776	YTPTSPN	Nonconsensus				

n represents the heptamer number; **Start** represents the position of the first amino acid in the heptamer. *Two of the heptamers include additional residues (heptamer n=2 features an extra Gly, and heptamer n=49 an extra Gly-Ser-Thr). In both cases, none of these extra amino acid residues include PTMs[37–41]. The color code corresponds to the following PTM classes: ■ unmodified, ■ Methylation, ■ Phosphorylation, ■ cis / trans isomerization, ■ Acetylation / Dimethylation, ■ O-GlcNAcylation / Phosphorylation, ■ Acetylation / Dimethylation / Methylation, ■ Asymmetric dimethylation / Citrullination / Dimethylation / Methylation / Symmetric dimethylation, ■ Acetylation

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/ Dimethylation / Methylation / SUMOylation / Trimethylation / Ubiquitination (Reproducibility Module 11, Supplementary Information S4).

Quantitatively, the total CTD state-space size can be computed as:

$$N_{\mathrm{CTD}} = \prod_{r=1}^{52+t} \left(\prod_{p=1}^{7} s_{r,p} \right),$$

where $s_{r,p}$ is the number of discrete states assigned to position p of repeat r according to the residue identity at that position (e.g., $s = 2$ for Tyr1; and $s = 4$ for the single Arg7; t corresponds to the 10-amino-acid residue C-terminal sequence.

Using the 52 heptads from UniProt P24928 (Table 1) plus the 10-residue C-terminal tail, and applying this residue-exact state mapping (Table 2) yields a CTD state-space size corresponding to:

$$N_{CTD} = 2_{Tyr1}^{22} \cdot 2_{Ser2}^{27} \cdot 3_{Ser2}^{3} \cdot 2_{Ser4}^{3} \cdot 2_{Ser5}^{36} \cdot 3_{Ser5}^{4} \cdot 3_{Ser7}^{1} \cdot 2_{Ser7}^{4} \cdot 2_{Ser9}^{1} \cdot 2_{Pro3}^{26} \cdot 2_{Pro6}^{25} \cdot 2_{Thr2}^{5} \cdot 3_{Thr4}^{2} \cdot 2_{Thr4}^{15} \cdot 2_{Thr5}^{1} \cdot 2_{Thr7}^{5} \cdot 3_{Thr7}^{1} \cdot 2_{Thr10}^{1} \cdot 4_{Lys7}^{5} \cdot 7_{Lys7}^{2} \cdot 3_{Lys7}^{1} \cdot 2_{Arg4}^{1} \cdot 6_{Arg7}^{1} \cdot 2_{Ser2}^{1} \cdot 2_{Ser6}^{1} \approx 3.83 \times 10^{63}$$

$log_{10}\,(N_{CTD}) \approx 63.58$, i.e., $N_{CTD} \sim 10^{64}$

This state-space size (Reproducibility Module 12, Supplementary Information S4) is conservative with respect to richer multi-mark models but remains an upper-bound capacity relative to in vivo occupancy distributions. Furthermore, PTMs on the remaining Pol II subunits ($Rpb1_{core}$–Rpb12) were not included, which would increase the theoretical state space by several additional orders of magnitude. Even under these assumptions, the combinatorial capacity of the CTD is enormous, encoding a combinatorial state space on the order of 10^{64}. In vivo, only a biologically constrained subset is kinetically accessible at any given time due to transcription-cycle ordering, enzymatic coupling, and non-uniform state occupancies. Nevertheless, the magnitude of the CTD's potential state space is consistent with its role as a dynamic regulatory scaffold capable of integrating diverse cellular signals and coordinating transitions across the transcription cycle.

II. Maximal information capacity of CTD PTM patterns.

To quantify the maximal information capacity of CTD we use Shannon's measure of state entropy[62]:

$$\mathrm{I_{CTD}} = \sum_i \log_2 (s_i)$$

Taking into account the number of discrete states (s_i), determined by the corresponding PTMs, the residue-level information, and considering all 52 heptapeptide repeats and the C-terminal tail (Table 2), the maximal information capacity in a human P24928 CTD yields the explicit sum:

$$\begin{aligned} I_{CTD} = {} & 22 \cdot \log_2(2) + 27 \cdot \log_2(2) + 3 \cdot \log_2(3) + 3 \cdot \log_2(2) + 36 \cdot \log_2(2) + 4 \cdot \log_2(3) \\ & + 1 \cdot \log_2(3) + 4 \cdot \log_2(2) + 1 \cdot \log_2(2) + 26 \cdot \log_2(2) + 25 \cdot \log_2(2) + 5 \cdot \log_2(2) \\ & + 2 \cdot \log_2(3) + 15 \cdot \log_2(2) + 1 \cdot \log_2(2) + 5 \cdot \log_2(2) + 1 \cdot \log_2(3) + 1 \cdot \log_2(2) \\ & + 5 \cdot \log_2(4) + 2 \cdot \log_2(7) + 1 \cdot \log_2(3) + 1 \cdot \log_2(2) + 1 \cdot \log_2(6) + 2 \cdot \log_2(2) \\ & \approx 211.22 \text{ bits} \end{aligned}$$

This value (Reproducibility Module 3, Supplementary Information 4) represents the upper bound of the theoretical information encoded by the known CTD PTMs, assuming independence among sites and equal probability across available states. In vivo, state probabilities are highly skewed and constrained

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by transcriptional stage, enzyme activity, and chromatin context; thus, the achieved entropy is expected to be lower. Nevertheless, the CTD functions as a high-dimensional regulatory medium capable of encoding complex cellular states through diverse PTM combinations (Figure 1).

Table 2. Residue-level information and contribution per PTM of the CTD (Rpb1 subunit)

Residue/site class	PTM state model	Count	States per site (s_i)	Bits/site $\log_2(s_i)$	Total bits
Tyr1	unmodified / phosphorylated	22	2	1.00	22.00
Ser2	unmodified / phosphorylated	27	2	1.00	27.00
Ser2	unmodified / O-GlcNAcylated / phosphorylated	3	3	1.58	4.75
Ser4	unmodified / phosphorylated	3	2	1.00	3.00
Ser5	unmodified / phosphorylated	36	2	1.00	36.00
Ser5	unmodified / O-GlcNAcylated / phosphorylated	4	3	1.58	6.34
Ser7 (1621)	unmodified / O-GlcNAcylated / phosphorylated	1	3	1.58	1.58
Ser7	unmodified / phosphorylated	4	2	1.00	4.00
Ser9 (heptad 49) (1938)	unmodified / phosphorylated	1	2	1.00	1.00
Pro3 (YSPTSPS, YSPSSPS & YSPTSPN)	cis / trans isomerization	26	2	1.00	26.00
Pro6 (YSPTSPS & YSPTSPN)	cis / trans isomerization	25	2	1.00	25.00
Thr2	unmodified / phosphorylated	5	2	1.00	5.00
Thr4	unmodified / O-GlcNAcylated / phosphorylated	2	3	1.58	3.17
Thr4	unmodified / phosphorylated	15	2	1.00	15.00
Thr5 (1885)	unmodified / phosphorylated	1	2	1.00	1.00
Thr7	unmodified / phosphorylated	5	2	1.00	5.00
Thr7 (1894)	unmodified / O-GlcNAcylated / phosphorylated	1	3	1.58	1.58
Thr10 (heptad 49) (1939)	unmodified / phosphorylated	1	2	1.00	1.00
Lys7	unmodified / acetylated / dimethylated / monomethylated	5	4	2.00	10.00
Lys7	unmodified / acetylated / dimethylated / monomethylated / SUMOylated / trimethylated / ubiquitinated	2	7	2.81	5.61
Lys7 (1908)	unmodified / acetylated / dimethylated	1	3	1.58	1.58
Arg4 (1603)	unmodified / monomethylated	1	2	1.00	1.00
Arg7 (1810)	unmodified / asymmetrically dimethylated / citrullinated / dimethylated / monomethylated / symmetrically dimethylated	1	6	2.58	2.58

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Residue/site class	PTM state model	Count	States per site (s_i)	Bits/site $\log_2(s_i)$	Total bits
Ser2, Ser6 C-terminal tail	unmodified / phosphorylated	2	2	1.00	2.00

Counts in Table 2 refer to residue instances for which a PTM state was included in the curated CTD model; residues present in the CTD sequence but lacking curated PTM evidence were assigned a single unmodified state and therefore contribute 0 bits. $I_{CTD} = \sum_i \log_2(s_i) = 211.22$ bits. (Reproducibility Module 13, Supplementary Information S4).

III. Information capacity of PTMs on the Rpb1 core (non-CTD)

Because Rpb1 PTM biology is not confined to the CTD, we separately quantified the human non-CTD Rpb1 PTM sites from UniProt, dbPTM, iPTMnet, PhosphoSitePlus and GlyGen (POLR2A / P24928), across residues 1–1592 (i.e., excluding the heptad-repeat CTD region to avoid double counting). These sites include acetylation, glutathionylation, methylation, O-GlcNAcylation, phosphorylation, sulfoxidation, SUMOylation and ubiquitination annotations (see Methods Section and Supplementary Table S1).

Using the same mutually exclusive state model, this conservative non-CTD set (208 PTMs across 159 unique modified residues) yields (Reproducibility Module 5, Supplementary Information 4):

$$I_{\text{Rpb1- core}} \approx 185.87 \text{ bits per Rpb1}$$

IV. Information capacity encoded by PTMs in the subunits Rpb2–Rpb12.

Although the CTD is the primary PTM-rich region of Pol II, structural subunits Rpb2–Rpb12 also carry numerous functionally relevant covalent modifications. Using human entries from UniProt, dbPTM, iPTMnet, PhosphoSitePlus and GlyGen, we compiled 278 residues modified by 338 unique PTM marks including acetylation, methylation, O-GlcNAcylation, phosphorylation, sulfoxidation, SUMOylation and ubiquitination. For each modified residue, and these individual PTM types the number of equally probable discrete states is $s_i = 2$. (Supplementary Tables S2–S12). When multiple PTM types act on the same residue we treat modifications as mutually exclusive (e.g., lysines modified by either ubiquitin or SUMO). Then, the composite state counts were calculated as:

$$s_{\text{composite}} = 1 + \sum_{j=1}^{k} (s_j - 1)$$

where s_j is the number of states for PTM type j at that residue. For example, a lysine capable of ubiquitination (s= 2) or SUMOylation (s= 2) yields $s_{\text{composite}}$ = 3, corresponding to Shannon information $\log_2(3) \approx 1.58$ bits.

The compiled dataset of Rpb2–Rpb12 PTM sites (Supplementary Tables S2-S12) shows the following aggregate contributions: phosphorylation sites in multiple serine, threonine, and tyrosine residues across subunits, each of them with s=2 unless part of a composite site. Acetylation sites in Alanines, Serines, lysines and N-terminal methionines, with s=2 unless in composite. Ubiquitination sites in lysines, each with s=2, except when sharing a residue with another PTM type. SUMOylation sites in lysines, with s=2, likewise adjusted in composites. Additionally, sulfoxidation is observed on methionine residues (three sites) and O-GlcNAcylation on one threonine residue, each with s=2. Finally, methylation

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sites in cysteines, lysines, methionines and arginines, with s=2. Note that virtually all composites involve lysines combining ubiquitination with one or more of SUMOylation, acetylation or methylation, giving s=3 or s=4. The one exception is S2 in Rpb6, which carries acetylation plus phosphorylation (s=3). After consolidating multi-PTM residues into composite sites the maximum theoretical information content in Rpb2–Rpb12 residues yields:

Rpb2: ~140.36 bits
Rpb3: ~33.51 bits
Rpb4: 10.00 bits
Rpb5: ~25.34 bits
Rpb6: ~5.58 bits
Rpb7: ~18.17 bits
Rpb8: ~26.58 bits
Rpb9: 15.00 bits
Rpb10: 8.00 bits
Rpb11: ~25.34 bits
Rpb12: 3.00 bits

$I_{Rpb2-Rpb12}$ = 310.89 bits per molecule

This information is equivalent to ≈ 38.86 bytes, per all Rpb2–Rpb12 complex (Reproducibility Module 14, Supplementary Information S4).

V. Total information capacity per Pol II molecule

Summing the three components (CTD + Rpb1 core + Rpb2–Rpb12): the combined information capacity of one human RNA Pol II molecule is:

$$I_{Pol\ II} = I_{CTD\ (Rpb1)} + I_{Rpb1\ core} + I_{Rpb2\text{-}Rpb12} \approx 211.22 + 185.87 + 310.89 \approx 707.98\ (\sim 88.50\ \text{bytes})$$

Thus, the maximal theoretical information capacity encoded by known PTM patterns on each human Pol II molecule corresponds to ~88.50 bytes of regulatory information (see Figure 1), distributed across dynamic and reversible covalent marks on Rpb1–Rpb12 (Reproducibility Module 15, Supplementary Information S4). This theoretical molecular information layer is reconfigurable and may support complex modulation of transcription through reversible chemical modifications.

VI. Cellular level PTM information capacity of Pol II.

Using an estimate of M ≈ 80,200 Pol II molecules per mammalian tissue-culture nucleus (experimental studies based on super-resolution microscopy, BioNumbers 112321[63]) the total information capacity at cellular level is:

I_{cell} = $I_{PoI\ II}$ x M ≈ 707.98 × 80,200 ≈ 5.68 × 10^7 bits ≈ 7.10 × 10^6 bytes ≈ 7.10 MB

This estimate (Reproducibility Module 15, Supplementary Information S4) represents a theoretical upper bound, derived from known PTMs assuming independence between sites, equal probability between states, and equal weighting of Pol II molecules. However, the accessible state space is strongly constrained by: transcriptional stage–specific PTM covalent marks, structural coupling between them, enzyme pathway wiring (kinases, phosphatases, CTD readers, chromatin state and transcriptional context). Thus, while Pol II carries a theoretical regulatory capacity on the order of a few megabytes,

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only a biologically meaningful subspace is used "in vivo". Here we report state-space information capacity (i.e., $\log_2(N)$ under an independent-site, equal-probability model), not the achieved Shannon entropy of "in vivo" PTM occupancy distributions, which is expected to be lower due to non-uniform occupancies and site coupling (see Supplementary Information S2). Nevertheless, this quantitative framework highlights the remarkable potential of PTM patterning to encode regulatory states at the molecular scale. In addition, sensitivity analysis shows that the high-dimensional Pol II PTM information capacity is robust to plausible alternative discrete-state model choices (Supplementary Information S3).

Figure 1

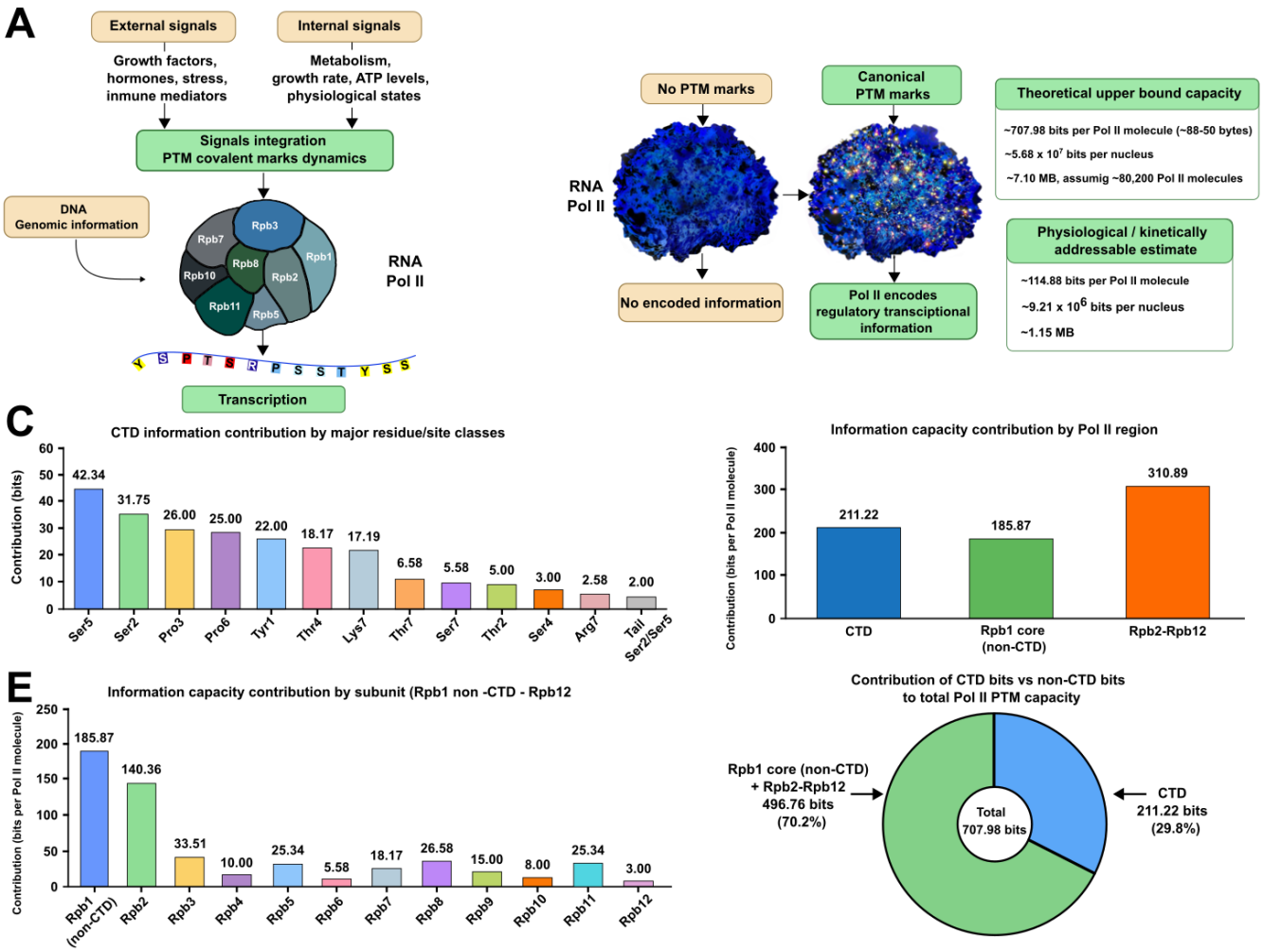


Figure 1. Information capacity encoded by post-translational modification patterning across human RNA polymerase II.
A, Schematic representation of external and internal cellular signals converging on RNA polymerase II (Pol II) through dynamic post-translational modification (PTM) marks. PTM writers, erasers, and readers convert environmental and intracellular cues into reversible covalent marks that modulate Pol II function during transcription. B, Conceptual comparison between an unmodified Pol II molecule and a canonically PTM-decorated Pol II molecule. In the absence of PTM marks, no PTM-encoded regulatory information is represented, whereas canonical PTM marks define a high-dimensional regulatory state space. The theoretical upper-bound capacity of the full human Pol II PTM landscape is estimated at ~707.98 bits per Pol II molecule, equivalent to ~88.50 bytes, ~5.68 × 10^7 bits per nucleus, or ~7.10 MB assuming ~80,200 Pol II molecules per nucleus. The conservative kinetically addressable estimate is ~114.88 bits per Pol II molecule, corresponding to ~9.21 × 10^6 bits per nucleus or ~1.15 MB. C, Contribution of major CTD residue/site classes to CTD information capacity. Values represent summed residue-class contributions under the curated discrete-state model. D, Regional contribution to total Pol II PTM information capacity: CTD, Rpb1 core non-CTD, and Rpb2–Rpb12. E, Per-subunit contribution of Rpb1 core non-CTD and Rpb2–Rpb12 to the non-CTD

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PTM information capacity. F, Relative contribution of CTD and non-CTD regions to the total theoretical Pol II PTM capacity. CTD contributes 211.22 bits, whereas Rpb1 core non-CTD plus Rpb2–Rpb12 contribute 496.76 bits, yielding a total of 707.98 bits per Pol II molecule.

VII. Pol II PTMs form a multi-timescale biochemical functional architecture.

Beyond static capacity, Pol II encodes regulatory information not only through the diversity of PTMs but also via the distinct kinetic lifetimes of their covalent marks and reversal mechanisms, generating a multi-timescale biochemical architecture (Figure 2). In fact, different PTM classes operate over fast, intermediate, and slow time scales, allowing the enzyme to encode and integrate signaling history over different temporal windows. This physiological information bound can be computed as:

$$I_{phys} = \log_2(N_{fast}) + \log_2(N_{Intermediate}) + \log_2(N_{slow})$$

For each kinetic layer k, we define:

$$N_k = a_k \hat{} m_k$$

where m_k is the number of effective sites or repeats and a_k is the effective number of kinetically distinguishable states per site after accounting for coupling and kinetic constraints. Therefore:

$$I_k = \log_2(N_k) = m_k \log_2(a_k)$$

VII-I. Fast-timescale layer (seconds–minutes)

Phosphorylation of the C-terminal domain at Tyr1, Ser2, Ser5, and Ser7 residues, mediates rapid transcriptional responses[64–66]. Kinetic studies indicate that phosphorylation and dephosphorylation cycles occur on the order of ~5 seconds to up to >15 min[45,46]. Phosphorylatable CTD heptads have an average ~3.12 phosphorylated residues (Table 1) that can be either unmodified or phosphorylated[67] (2^3 = 8 states per heptad). Although the canonical CTD sequence contains 52 heptads, CTD mass spectrometry in human cells shows sparse phosphorylation[67,68], with stoichiometry studies in human CTD estimating ~30 phosphorylated heptads per CTD[69]. However, the effective distinguishable states are further reduced due to kinetic coupling and electrostatic and steric constraints, specifically to ~5 per heptad (unmodified state, ~3 distinct monophosphorylated states and a rarer double phosphorylated state)[66,67,70,71].

We thus selected $m_{fast} \approx 30$ modified heptads and $a_{fast} \approx 5$ as a conservative effective-state value within the restricted kinetically distinguishable configurations per active heptad. The Shannon information is:

$$I_{fast} = \log_2(5\hat{}30) = 30 \log_2(5) \approx 69.66 \text{ bits}$$

Therefore, the physiological information capacity of this layer is approximately 70 bits per Pol II molecule (Figure 2), which is equivalent to $5\hat{}30 \approx 9.31 \times 10\hat{}20$ distinguishable rapid-response CTD states (Reproducibility Module 16, Supplementary Information S4). This short transient information enables Pol II to give dynamically fast responses to extracellular cues and numerous intracellular variables, coordinating transcription initiation, elongation, and promoter-proximal pausing in real time.

VII-II. Intermediate layer (minutes to hours).

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Biochemical time-course kinetic assays indicated a number of 21 Pro6 isomerization processes along the Ser5-Pro6 bonds of consensus heptad sequences (YSPT**S-P**S), in the human CTD[37], which spans the scale of ~30 seconds (catalyzed) to up to ~30 min (uncatalyzed)[47,48]. Therefore we have $m_{Pro6} \approx 21$ and $a_{Pro6} = 2$. Likewise, kinetic and functional studies show that O-GlcNAcylation addition and removal cycles occur within < 2min to ~30 min[49–51]. PTM databases reported nine O-GlcNAc marks in the human CTD (eight by GlyGen, and one by PhosphoSitePlus) and two more are reported by experimental works with human CTD using LC-MS/MS[49]. Therefore, $m_{O\text{-}GlcNAc} \approx 11$, with $a_{O\text{-}GlcNAc} = 2$.

Thus, for Pro6 isomerization and O-GlcNAcylation processes we obtain a number of effectively modified sites of $m_{Intermediate} = 32$, and $a_{Intermediate} = 2$ kinetically distinguishable states per site. The information contained in both processes according to Shannon information theory is:

$$I_{Intermediate} = 32 \cdot log_2(2) \approx 32\ bits$$

This mesoscale physiological information corresponds to $N_{intermediate} \approx 2^{32} \approx 4.29 \times 10^9$ possible functional configuration states (Reproducibility Module 16, Supplementary Information S4). The Pol II mesoscale information represents a notable reservoir of mid-term regulatory history (Figure 2). Intermediate PTM modifications integrate metabolic signals, chromatin context, and cofactor interactions, providing a transient regulatory layer that sustains transcriptional adaptations over tens of minutes, bridging rapid signaling events and mesoscale-term regulatory processes.

VII-III. Slow layer (hours–days).

Rpb1 ubiquitination, including K48- and K63-linked ubiquitin chains on Rpb1 K1268, coordinates Pol II degradation, repair, and transcription recovery after DNA damage[24,72]. Ubiquitination establishes a slower transient process that can persist over longer windows, from ~15 min to ~24 hours, and influences the available Pol II pool[24,52,53]. This slower timescale encodes information about long-term cellular perturbations such as DNA damage, oxidative stress, and arrested or defective Pol II states, influencing the pool of available Pol II molecules and thereby shaping transcriptional output over extended periods[24,73,74]. Experimentally, across the entire 12-subunit human RNA Pol II complex proteomics/PTM resources report on the order of ~120 distinct ubiquitinated residues in total[37–41], but only a biologically constrained subset is kinetically accessible at any given time under physiological conditions. Assuming a very conservative estimation for ubiquitination, of m_{slow} ~10 slow sites/processes, with an average a_{slow} ~2.5 distinguishable states per site (capturing long-lived outcomes such as stable ubiquitination-linked states) the Shannon information is:

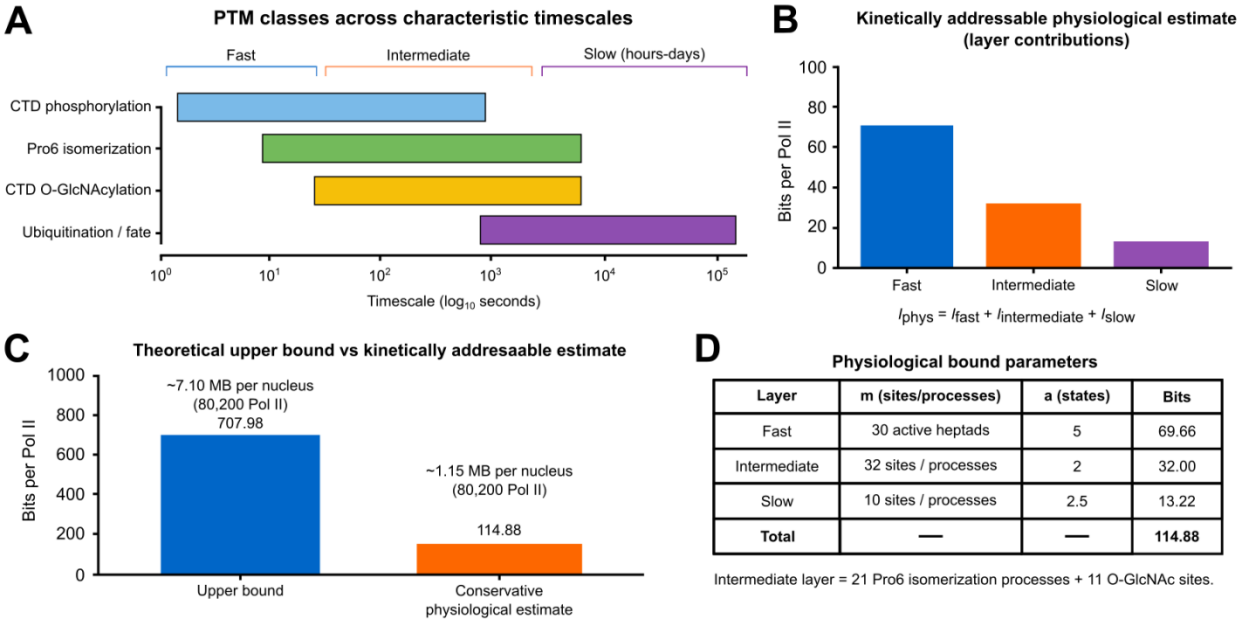
$$I_{slow} \approx 10 \log_2 (2.5) \approx 13.22 \text{ bits}$$

This physiological slow process encodes roughly 13.2 bits per Pol II molecule, corresponding to a potential space of ~9.54 × 10^3 (2.5^10) of functional configuration states. This layer is smaller than the rapid one (~70 bits) but extremely powerful because it persists for up to ~24 h influencing Pol II stability and genome-wide transcriptional readiness (Reproducibility Module 16, Supplementary Information S4).

Timescale marks such as acetylation, SUMOylation, and methylation are not explicitly parameterized here because current condition-resolved kinetic occupancy experimental data are insufficient to define a defensible effective state model.

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Figure 2



Layer	m (sites/processes)	a (states)	Bits
Fast	30 active heptads	5	69.66
Intermediate	32 sites / processes	2	32.00
Slow	10 sites / processes	2.5	13.22
Total	—	—	**114.88**

Figure 2. Multi-timescale physiological architecture of Pol II PTM information.
A, Characteristic kinetic regimes of major Pol II PTM classes included in the physiologically addressable model. CTD phosphorylation operates predominantly on fast seconds-to-minutes timescales, Pro6 cis/trans isomerization and CTD O-GlcNAcylation define an intermediate minutes-scale layer, and ubiquitination-linked Pol II fate decisions extend into slower hours-to-days regulatory windows. B, Layer-wise contribution to the conservative kinetically addressable physiological estimate. The fast layer contributes 69.66 bits per Pol II molecule, the intermediate layer contributes 32.00 bits, and the slow layer contributes 13.22 bits, giving a total of 114.88 bits per Pol II molecule. C, Comparison between the theoretical upper-bound PTM state-space capacity and the conservative physiological estimate. The theoretical upper bound is 707.98 bits per Pol II molecule, corresponding to ~7.10 MB per nucleus assuming ~80,200 Pol II molecules, whereas the conservative physiological estimate is 114.88 bits per Pol II molecule, corresponding to ~1.15 MB per nucleus. D, Parameters used for the coarse-grained physiological estimate. The fast layer was modeled as 30 active heptads with 5 kinetically distinguishable states per heptad; the intermediate layer as 32 sites/processes with 2 states each; and the slow layer as 10 sites/processes with an average of 2.5 distinguishable states. The intermediate layer comprises 21 Pro6 isomerization processes and 11 CTD O-GlcNAc sites.

VII-IV. Integrative Pol II physiological multi-timescale architecture

Combining these timescales, Pol II PTMs establish a biochemical dynamic structure, from transient second-scale responses to longer-lived hours-to-days regulatory states. This hierarchy enables the cell to integrate recent transcriptional events with transient physiological trends. Thus, the conservative kinetically addressable estimate is:

$$I_{phys} \approx I_{fast} + I_{intermediate} + I_{slow} \approx 69.66 + 32 + 13.22 \approx 114.88 \text{ bits per Pol II molecule}$$

This physiological value (Reproducibility Module 16, Supplementary Information S4) is a conservative estimate of kinetically addressable information, distinct from the theoretical upper bound derived from the full ~707.98 bits residue-level state space (Results: Section V). The remaining ~593.10 bits (707.98 – 114.88 bits) reflect static combinatorial potential not fully expressed simultaneously due to kinetic constraints (Figure 2). Only a subset of PTMs operates in any given temporal window, therefore, an individual Pol II molecule cannot biologically sample all theoretical states in real time. Nevertheless, 114.88 bits of kinetically accessible, timescale-separated information is extraordinarily notable for a single protein.

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At M = 80,200 Pol II molecules per nucleus (BioNumbers 112321[63]):

$$I_{phys,\ cell} \approx 114.88 \times 80{,}200 \approx 9.21 \times 10\text{^}6 \text{ bits} \approx 1.15 \text{ MB}$$

This physiological bound is kinetically addressable information, which represents a remarkable amount of physiological transcriptional value at the cellular level, distinct from the theoretical upper bound derived from the full residue-level state space. Together, these experimentally motivated estimates provide a quantitative framework that reframes Pol II PTM patterning as a high-dimensional, dynamically reconfigurable, multi-timescale information-processing system.

Figure 3

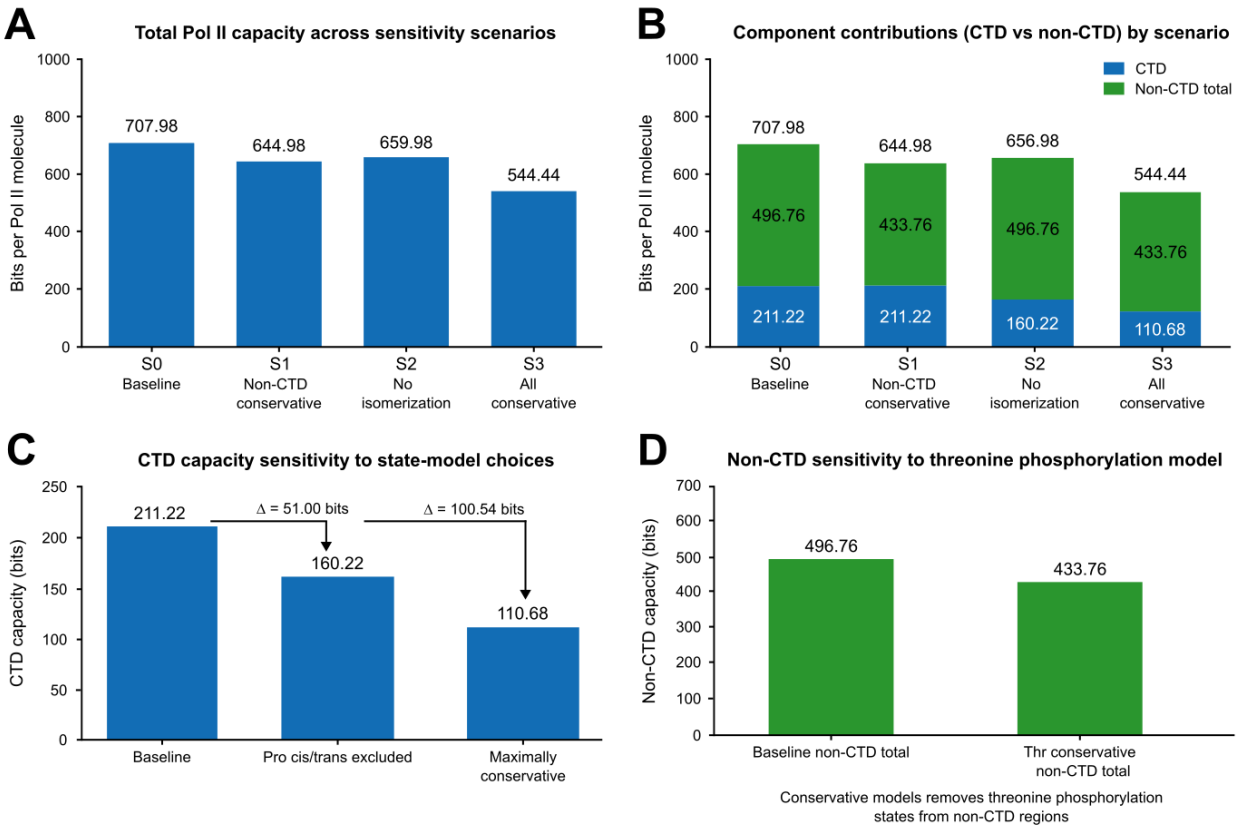


Figure 3. Sensitivity analysis of Pol II PTM information-capacity estimates.
A, Total Pol II theoretical information capacity across four sensitivity scenarios. The baseline model (S0) yields 707.98 bits per Pol II molecule. A non-CTD conservative model removing threonine phosphorylation states from non-CTD regions (S1) yields 644.98 bits. A CTD model excluding Pro cis/trans isomerization (S2) yields 656.98 bits. The fully conservative model combining non-CTD threonine-phosphorylation removal with maximal CTD conservative assumptions (S3) yields 544.44 bits. B, Component contributions of CTD and non-CTD regions across the same scenarios. CTD contributions are shown separately from the combined non-CTD total, corresponding to Rpb1 core non-CTD plus Rpb2–Rpb12. C, Sensitivity of CTD capacity to alternative state-model choices. Excluding Pro cis/trans isomerization reduces CTD capacity from 211.22 to 160.22 bits, a reduction of 51.00 bits. The maximally conservative CTD model reduces CTD capacity to 110.68 bits, a reduction of 100.54 bits relative to baseline. D, Sensitivity of the non-CTD capacity to a conservative threonine-phosphorylation model. Removing threonine phosphorylation states from non-CTD regions reduces the combined non-CTD capacity from 496.76 to 433.76 bits. Together, these sensitivity analyses show that the conclusion of a high-dimensional Pol II PTM information landscape remains robust under conservative alternative state-model assumptions.

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Discussion

Pol II activity is central to human cellular regulation, helping ensure that only the specific genomic information required for optimal physiological processes is transcribed at any given moment of the cell cycle.

This macromolecular enzyme operates within a complex regulatory network that is tuned at multiple stages by transcription factors, activators and repressors, coactivators, and chromatin-modifying enzymes. These regulators interact in complex ways with Pol II to ensure that specific genomic information is expressed at the right time, in the appropriate amounts, and in any particular cell type, in response to diverse environmental and intracellular signals[36].

A fundamental feature of Pol II is its dense layer of more than 775 PTMs, which together form a dynamic molecular marking system that integrates a great number of cellular signaling pathways and physiological cues modulating transcription initiation, elongation, and termination. These PTMs, most prominently phosphorylation, acetylation, ubiquitination, glycosylation, methylation, isomerization, citrullination, and SUMOylation, act as a tightly regulated, reversible control layer that coordinates transcription through combinatorial patterns[36,75]. Although the genomic information content of the cell has been well-documented (ENCODE Project Consortium 2012[42]), the potential information capacity associated with covalent marks of Pol II remains largely unexplored.

In this work, we provide a quantitative framework for understanding Pol II PTM patterning as a distributed regulatory information layer across the full enzyme (Rpb1–Rpb12). Under a discrete-state model and Shannon state-entropy assumptions (independence and equal state probabilities), we first estimate a theoretical maximum information capacity of ~707.98 bits per Pol II molecule. At the cellular level, assuming ~80,200 Pol II molecules per nucleus, this corresponds to ~5.68 × 10^7 bits (~7.10 MB). These values are explicit upper bounds and do not imply that all states are sampled, occupied, or biologically accessible simultaneously in vivo. PTM occupancies are constrained by transcriptional stage, enzymatic wiring (writers/erasers/readers), and structural coupling among sites, so only a subset of PTMs is active within any given temporal window.

Consistent with these constraints we estimated a conservative kinetically addressable value of ~114.88 bits per molecule (~1.15 MB per nucleus). This distinction is essential since the theoretical capacity reflects the size of the state space defined by known covalent marks, whereas the physiological capacity reflects only what can plausibly be written, maintained, and read given kinetic and structural constraints.

The results also emphasize the vast combinatorial potential of Pol II PTM patterning. For the CTD alone, the modeled PTM pattern space is in the order of $\sim 10^{64}$ possible configurations, illustrating the scale of the regulatory landscape even under conservative mutual-exclusivity assumptions. While cells sample only a small and biologically meaningful subset of this space, the size of the underlying state space helps contextualize why PTM patterning can support rich and adaptable regulation.

Importantly, our physiological and kinetic analysis also suggests that Pol II encodes regulatory information not only through the combinatorial diversity of PTMs forming a flexible mark pattern code[76]. Enzymes responsible for covalent modification marks operate over distinct timescales constituting a multilayered molecular system that enables Pol II to encode the recent transcriptional and signaling history in a temporally ordered manner through distinct kinetic activities. In fact, the PTMs involved in regulatory transcriptional processes exhibit distinct lifetimes, forming fast (~5 seconds–minutes), intermediate (~1–30 min) and slow (~15 min – ~24h) biochemical transient processes. Rapid

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phosphorylation cycling supports second-to-minute responsiveness, whereas intermediate marks such as proline isomerization and O-GlcNAcylation integrate signals over longer windows and slower ubiquitination-linked processes can encode more persistent regulatory history through polymerase recycling and turnover. This kinetic-layer analysis yields a conservative cell-level estimate of ~1.15 MB of kinetically addressable regulatory state capacity, distinct from the theoretical maximum.

Finally, we addressed how non-uniform state probabilities and coupling reduce achieved entropy relative to state-space capacity (Supplementary Information S2). We also performed sensitivity analyses (Figure 3) showing that the conclusion of high-dimensional PTM pattern capacity is robust to plausible alternative discrete-state levels for specific PTM classes (Supplementary Information S3). A more precise estimation of achieved entropy would require (i) condition-resolved occupancy distributions, (ii) joint correlations among sites on the same Pol II molecule, and (iii) sufficient single-molecule resolution to distinguish combinatorial patterns rather than population averages. These requirements remain experimentally challenging; therefore, a more precise entropy estimation of Pol II is outside the scope of the present work. All numerical results shown here are fully reproducible using the "Python Reproducibility Modules" provided in Supplementary Information S4.

Overall, this quantification, grounded in Shannon's information theory, reframes Pol II PTM landscapes as a distributed, dynamically reconfigurable regulatory layer whose capacity can be estimated transparently and refined as richer occupancy and correlation data become available. Similarly, our study highlights the quantitative scale of the functional coupling between genetic information and cellular physiological processes. In a broader sense, casting PTM landscapes in information-theoretic terms provides an analytical language to connect biochemical complexity with regulatory potential, and offers a quantitative basis for considering how dysregulation of PTM writers, erasers, and readers could alter transcriptional control and contribute to profound cellular consequences.

Acknowledgments

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Declaration of interests

The authors declare that they have no competing interests.

Author contributions

IMDF: conceived, designed and directed the investigation, writing – original draft. MF, IM, LM, JC-P, BC-P, CB, GPY, JMC and JIL: investigation, research mapping, writing – review and editing. JC-P and BC-P Python Modules. All authors wrote the manuscript and agreed with its submission.

Data availability

The version-pinned UniProt sequences used for residue mapping are provided as Supplementary Data File 3 (uniprot_polII_seqs_2026_05_22.tsv). The integrated RNA Polymerase II PTM atlas used in this study is provided as Supplementary Data File 4 (ptms_polII_2026_05_22.tsv). Both supplementary data files are also deposited at https://doi.org/10.5281/zenodo.20345702. All source data were retrieved from six publicly available databases: UniProt UniSave (https://www.ebi.ac.uk/uniprot/unisave/app/#/), UniProtKB (https://www.uniprot.org/uniprotkb/), EBI Proteins (https://www.ebi.ac.uk/proteins/api/), dbPTM (https://biomics.lab.nycu.edu.tw/dbPTM/), iPTMnet

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(https://research.bioinformatics.udel.edu/iptmnet/), PhosphoSitePlus (https://www.phosphosite.org/), and GlyGen (https://glygen.org/). The estimate of 80,200 RNA Polymerase II copies per mammalian nucleus was taken from BioNumbers (ID 112321; Zhao et al. 2014[63]; https://bionumbers.hms.harvard.edu/bionumber.aspx?s=n&v=0&id=112321).
Literature-mined PTM entries were incorporated from Harding et al. (1992)[59], Noble et al. (2005)[60], Xiang et al. (2010)[56], Werner-Allen et al. (2010)[47], Zhang et al. (2012)[48], Mayfield et al. (2015)[57], Lewis et al. (2016)[49], Gibbs et al. (2017)[58] and Sharma et al. (2019)[61].

Code availability

The Jupyter notebooks required to fully reproduce all tables and numeric results from these source data are described in Supplementary Information S4 and provided as Supplementary Data File 1 and Supplementary Data File 2, and have also been deposited on GitHub (https://github.com/JoseCarrascoPujante/Multi-timescale_PTM_architecture_in_human_RNA_polymerase_II) and Zenodo (https://doi.org/10.5281/zenodo.20345702).

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STAR★Methods

We quantified the information capacity encoded by PTM patterns in human Pol II using an explicit discrete-state model at the residue level and computing Shannon maximal entropy under a uniform, independent-site assumption. We mainly report (I) a theoretical upper bound based on the full PTM state space and (II) a physiological/kinetically accessible estimate derived from coarse-graining PTM dynamics into multi-timescale layers.

Pol II subunit definitions and sequence sources

In our study, we considered the canonical 12-subunit human complex of Pol II (Rpb1–Rpb12; gene products POLR2A–POLR2L and associated subunits). Protein sequences, isoform identifiers, and residue coordinates were obtained from UniProt entries corresponding to the reviewed human proteins. Residue numbering was standardized to UniProt canonical sequences. For Rpb1, the CTD region and repeat numbering were defined using the annotated human CTD architecture (UniProt P24928[37]). PTM sites were compiled from curated PTM resources (UniProt, dbPTM, iPTMnet, PhosphoSitePlus and GlyGen; supplementary cross-checking with UniProt feature annotations and PhosphoSitePlus where relevant). For each subunit, all the supported PTM sites were extracted with residue identity, position, and PTM type.

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Input data.
PTM sites were imported from the free access resources of said online PTM databases considering their residue position, residue identity, PTM type(s), and UniProt accession. Residue coordinates were standardized to UniProt canonical sequences; for Rpb1, the CTD region and repeat boundaries were defined using the annotated POLR2A CTD architecture (52 heptads, UniProt P24928[37]).

Data cleaning and normalization.
PTM entries for each subunit were mapped to a canonical site key from the UniProt protein sequence (protein accession, residue letter, residue index). Duplicate entries across sources (same site and PTM type) were merged. For residues annotated with multiple PTM types, all PTM types were retained for that site. Ambiguous entries (e.g., unspecified residue within a peptide) and sites not mappable to the canonical sequence were excluded. CTD (Rpb1) heptad-repeat sites were modeled explicitly using mammal-relevant PTM classes (Figure 1). PTMs on residues outside the CTD were included in the Rpb1 core non-CTD set to avoid double counting.

Discrete PTM state models (per residue)
Each modifiable residue i was assigned a discrete number of mutually exclusive states s_i reflecting the PTM chemistry used in the model. Unless otherwise stated, "unmodified" was treated as one state, and modified forms were treated as additional mutually exclusive states.

CTD PTM model
For CTD heptads, residues were assigned:
-Tyr1: unmodified/phosphorylated (22 sites)
-Ser2: unmodified/phosphorylated (27 sites)
-Ser2: unmodified/O-GlcNAcylated/phosphorylated (3 sites)
-Ser4: unmodified/phosphorylated (3 sites)
-Ser5: unmodified/phosphorylated (36 sites)
-Ser5: unmodified/O-GlcNAcylated/phosphorylated (4 sites)
-Ser7 (1621): unmodified/O-GlcNAcylated/phosphorylated (1 site)
-Ser7: unmodified/phosphorylated (4 sites)
-Ser9 (heptad 49) (1938): unmodified/phosphorylated (1 site)
-Pro3 (YSPTSPS, YSPSSPS & YSPTSPN): cis/trans isomerization (26 sites)
-Pro6 (YSPTSPS & YSPTSPN): cis/trans isomerization (25 sites)
-Thr2: unmodified/phosphorylated (5 sites)
-Thr4: unmodified/O-GlcNAcylated/phosphorylated (2 sites)
-Thr4: unmodified/phosphorylated (15 sites)
-Thr5 (1885): unmodified/phosphorylated (1 site)
-Thr7: unmodified/phosphorylated (5 sites)
-Thr7 (1894): unmodified/O-GlcNAcylated/phosphorylated (1 site)
-Thr10 (heptad 49) (1939): unmodified/phosphorylated (1 site)
-Lys7: unmodified/acetylated/dimethylated/monomethylated (5 sites)
-Lys7: unmodified/acetylated/dimethylated/monomethylated/ SUMOylated/trimethylated/ubiquitinated (2 sites)
-Lys7 (1908): unmodified/acetylated/dimethylated (1 site)
-Arg4 (1603): unmodified/monomethylated (1 site)
-Arg7 (1810): unmodified/asymmetrically, dimethylated/citrullinated/dimethylated/monomethylated/symmetrically demethylated (1 site)
-Ser2, Ser6 C-terminal tail: unmodified/phosphorylated (2 sites)

The CTD was modeled as 21 canonical repeats with consensus **YSPTSPS** and 31 non-consensus repeats. Residues present in the CTD sequence but not assigned a curated PTM state in this model were treated as having $s_i = 1$ and therefore contributed 0 bits to the information capacity.

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Non-CTD PTM model (Rpb1 core and Rpb2–Rpb12)

Outside the CTD, PTMs were modeled using the following state sets:

- Acetylation: unmodified / acetylated (s = 2).
- Glutathionylation: unmodified / glutathionylated (s = 2).
- Methylation: unmodified / methylated (s = 2).
- O-GlcNAcylation: unmodified / O-GlcNAcylated (s = 2).
- Phosphorylation: unmodified / phosphorylated (s =2).
- Sulfoxidation: unmodified / sulfoxidized (s = 2).
- SUMOylation: unmodified / SUMOylated (s = 2).
- Ubiquitination: unmodified / ubiquitinated (s = 2).

Multi-PTM residues and composite state counts

If a residue was annotated with more than one PTM type (e.g., a serine that can be phosphorylated or O-GlcNAcylated), we treated PTM types as **mutually exclusive** at that residue and defined an effective composite state count:

$$s_{\text{composite}} = 1 + \sum_{j=1}^{k} (s_j - 1)$$

where s_j is the state count for PTM type j acting at that residue and the "1" represents the unmodified state shared across PTM types.

This choice avoids double-counting the unmodified state and should be interpreted as a conservative residue-level state model, not as a claim that simultaneous multi-mark occupancy cannot occur in all biological contexts.

Information calculations.

Per-site information was computed as $I_i = \log_2(s_i)$. For a set of n modeled sites:

$$I_{\max} = \sum_{i=1}^{n} \log_2 (s_i) = \log_2 \left(\prod_{i=1}^{n} s_i \right),$$

and the pattern-space size is $N = \prod_{i=1}^{n} s_i$ with $I_{\max} = \log_2(N)$. Total Pol II capacity was computed as:

$$I_{\text{PolII}} = I_{\text{CTD}} + I_{\text{Rpb1-core}} + I_{\text{Rpb2-Rpb12}}.$$

Cellular capacity was computed as $I_{\text{cell}} = M \cdot I_{\text{PolII}}$, with $M = 80{,}200$ Pol II molecules per nucleus.

Physiological/kinetically accessible estimate

To estimate a conservative physiologically accessible information capacity, PTMs were coarse-grained into three explicitly parameterized kinetic layers: a fast CTD phosphorylation layer, an intermediate Pro6 isomerization/O-GlcNAcylation layer, and a slow ubiquitination-linked layer. Acetylation, SUMOylation, and methylation were acknowledged qualitatively but not explicitly parameterized because current condition-resolved kinetic occupancy data are insufficient to define a defensible effective-state model.This physiological information bound can be computed as:

$$I_{\text{phys}} = \log_2 (N_{\text{fast}}) + \log_2 (N_{\text{intermediate}}) + \log_2 (N_{\text{slow}})$$

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For each layer k, $N_k = a_k \hat{} m_k$ where m_k is the effective number of sites/processes and a_k is the effective number of kinetically distinguishable states per site/process. Therefore $I_k = \log_2(N_k) = m_k \log_2(a_k)$

Reproducibility

Supplementary Information S4 provides a Python implementation that reproduces all numerical calculations and tables. The pipeline requires Python ≥3.10 together with pandas, requests, IPython/Jupyter utilities, and standard-library modules.

Limitations

The reported maximum theoretical capacities are upper bounds derived from a discrete-state model under independence and equal-probability assumptions. In vivo, many marks are rare, transient, or stage-restricted, and sites are not independent due to enzymatic coupling, reader-mediated cooperativity, steric/structural constraints, and transcription-cycle ordering, all of which reduce achieved entropy relative to the theoretical maximum (Supplementary Information S2). PTM catalogs are incomplete and biased toward abundant proteins and detectable chemistries, so the enumerated state space may omit true sites or include sites with negligible occupancy. Partial observability is also a major limitation: many measurements are population-averaged and do not resolve correlated site occupancies on individual Pol II molecules. More kinetic lifetime measurements and single-molecule PTM occupancy data will be required to estimate achieved entropy more precisely.

Supplementary Information S1. Pol II architecture

Human RNA polymerase II is organized as a 12-subunit complex, Rpb1–Rpb12, built around a catalytic core whose structural and mechanistic principles are conserved from archaea to humans[54]. The largest subunit, Rpb1, carries the extended C-terminal domain (CTD), a repetitive regulatory platform that is densely targeted by reversible PTMs. These CTD modifications contribute to the temporal coordination of Pol II activity across the transcription cycle, from initiation and promoter escape to productive elongation, termination, and co-transcriptional RNA processing[44]. Structural and functional studies of the Pol II pre-initiation complex (PIC) show that general transcription factors bind Pol II surfaces including Rpb2 domains (especially the protrusion and lobe domains) and the core initiation module which includes the Rpb3/Rpb11 heterodimer[77,78]. The Rpb4/Rpb7 heterodimer forms a specialized RNA-binding module[79,80] whose phosphorylation appears to regulate its engagement with RNA polymerase II[81–83], linking elongation and termination[81,84] to mRNA imprinting[81,83] and decay[80,83]. Rpb5 stability and its assembly into RNA Pol II, which influences nuclear Pol II levels[85], is promoted by URI[86], a transcriptional repressor[87,88] whose multiple phosphorylation and acetylation states position it as a potential regulatory hub translating extracellular signalling into specific patterns of RNA Pol II transcriptional activity[86]. Separately, phosphorylation of the Rpb6 N-terminal domain has been postulated to confer it with regulatory functions[89].

The small subunits Rpb8–Rpb12 also participate in PTM-dependent processes essential for Pol II function. It has been suggested that phosphorylation of the small subunits Rpb8, Rpb9, Rpb10 and Rpb12 may influence transcriptional activity[90]. Notably, Rpb9 has documented roles in initiation-factor interactions[91], transcriptional fidelity[92,93], transcription elongation rate[94] and transcription-coupled DNA repair[95]. In the cellular response to UV-induced DNA damage, non-proteolytic polyubiquitination of Rpb8 approximately doubles cell survival rates[96], while Rpb9-promoted ubiquitination and proteasomal degradation of subunit Rpb1 disassembles RNA Pol II molecules arrested at DNA lesions[97]. Rpb11 and Rpb12 are small Pol II subunits associated with the Rpb3-containing assembly subcomplex[98–100]. Rpb11 supports Pol II assembly through its heterodimerization with Rpb3[98,101] and participates in the Mediator-bound preinitiation complex[78]. It also carries four acetylation sites, two of which have been predicted to

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stabilize the Rpb3/Rpb11 interface[102]. In turn, Rpb12 contributes to Pol II biogenesis[99] and, based on ortholog evidence from *Archaea*, also fosters promoter opening[103]. It carries surface-exposed phosphosites with the potential to influence its interactions with transcription regulators[90]. Collectively, PTMs across the entire Pol II subunit ensemble (Rpb1–Rpb12), with over 630 amino acid residues annotated as undergoing over 775 covalent modifications (Reproducibility Module 17, Supplementary Information S4), are centered on Rpb1, whose CTD PTMs connect RNA Pol II to co-transcriptional RNA processing, termination and transcription-coupled DNA repair[44,104,105]. Beyond Rpb1, phosphorylation of the Rpb4/7 heterodimer modulates stalk stoichiometry and links transcription termination to mRNA imprinting[80,81], and Rpb8 ubiquitination regulates the DNA-damage response[96]. For the remaining subunits, evidence is still dominated by PTM mapping and indirect pathway effects, leaving RNA Pol II full-ensemble coordination unresolved[106,107].

Supplementary Information S2. From state-space capacity to achieved entropy under non-uniform and correlated PTM probabilities

S2.1. Purpose and interpretation

Our first results report a state-space capacity upper bound for Pol II PTM patterns under a discrete-state model in which each site i has s_i mutually exclusive states. Under site independence and uniform state probabilities, the maximal information capacity is:

$$I_{\max} = \sum_i \log_2 (s_i) = \log_2 \left(\prod_i s_i \right).$$

This quantity equals the Shannon entropy only when state probabilities are uniform at each site and sites are independent. The purpose of this Supplementary Information is to make this distinction explicit and to show, with simple examples, how non-uniform state probabilities and site coupling reduce achieved entropy relative to maximal state-space capacity.

Accordingly, throughout the manuscript, capacity refers to the logarithm of the modeled state-space size, whereas 'achieved entropy' refers to the entropy of an empirical occupancy distribution

S2.2. Site-wise entropy under non-uniform occupancies

If site i has state probabilities $\{p_{ik}\}_{k=1}^{s_i}$, the achieved (site-wise) entropy is:

$$\mathrm{H}_i = -\sum_{k=1}^{s_i} p_{ik} \log_2(p_{ik}),$$

and under independence the achieved entropy is $\mathrm{H} = \sum_i \mathrm{H}_i$. Because biological PTM occupancies are typically sparse, stage-dependent, and often dominated by the unmodified state, $\mathrm{H} \ll \mathrm{I}_{\max}$ in most regimes.

To show how non-uniform priors reduce achieved entropy, we consider simple distributions in which the unmodified state dominates. These distributions serve to demonstrate both scale and directionality.

S2.2.1. Binary sites (2 states): unmodified vs modified

For a binary PTM site with states unmodified and modified, let p be the probability that the site is modified. The state probabilities are therefore 1 − p and p. Then, the achieved entropy is:

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$$H_{binary}(p) = -(1-p)\log_2(1-p) - p\log_2(p)$$

Examples are shown in Table S2.1.

Modified probability p	Unmodified probability 1 - p	Achieved entropy H(p) (bits)	Fraction of binary maximum
0.50	0.50	1.000	100.0%
0.10	0.90	0.469	46.9%
0.01	0.99	0.081	8.1%

Even for binary PTMs, sparse occupancy can reduce effective bits per site by an order of magnitude relative to the maximum.

S2.2.2. Three-state sites (3 states): unmodified / phospho / O-GlcNAcylated

For a three-state site such as a serine or threonine modeled as unmodified, phosphorylated, or O-GlcNAcylated, the maximal state-space capacity is $\log_2(3) = 1.585$ bits. However, a non-uniform occupancy distribution gives a smaller achieved entropy.

$$\text{Example distribution: } p = (0.80, 0.15, 0.05)$$

then:

$$H_3 = -[0.80\log_2(0.80) + 0.15\log_2(0.15) + 0.05\log_2(0.05)] = 0.884 \text{ bits}$$

In this illustrative case, the achieved entropy is approximately 56% of the maximal three-state capacity. This example is not intended as a measured occupancy distribution; it simply demonstrates how non-uniform probabilities reduce entropy relative to state-space capacity.

S2.2.3. Four-state sites: arginine modification example

For a four-state site, such as an arginine modeled as unmodified, asymmetrically dimethylated, symmetrically dimethylated, or citrullinated, the maximal capacity is $\log_2(4) = 2$ bits. If the unmodified state dominates, the achieved entropy is again much lower.

$$\text{Example distribution: } p = (0.85, 0.05, 0.05, 0.05),$$

then:

$$H_4 = -[0.85\log_2(0.85) + 3 \times 0.05\log_2(0.05)] = 0.848 \text{ bits}$$

This illustrates that high nominal state granularity does not imply high achieved entropy unless the alternative states are substantially populated. The numerical occupancy examples above are illustrative and are not used to infer in vivo Pol II entropy.

S2.3. State-space capacity versus achieved entropy

The residue-level capacities reported in the main text quantify the size of the modeled PTM state space under a uniform independent-site assumption. This is useful because it provides an auditable upper bound and allows direct comparison among CTD, Rpb1-core, and Rpb2–Rpb12 contributions.

However, achieved in vivo entropy would require condition-resolved probabilities for every PTM state at every site. If the sites were independent and if all marginal probabilities were known, the achieved entropy would be:

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$$H_{achieved,\ independent} = \Sigma_i\ H_i$$

In practice, PTM state probabilities are likely to be non-uniform because many marks are rare, transient, condition-specific, or transcription-stage-specific. Therefore:

$$H_{achieved} \leq I_{max}$$

The inequality is expected to be strict for most biological contexts.

S2.4. Reduction of entropy by site coupling and correlations

Pol II PTMs are not generally independent. Shared writers and erasers, CTD-reader recruitment, steric constraints, structural coupling, transcription-cycle ordering, and chromatin context can generate statistical dependence among sites.

For two sites X and Y, the joint entropy satisfies:

$$H(X,Y) = H(X) + H(Y) - I(X;Y)$$

where I(X;Y) is the mutual information between the two sites. If the two sites are independent, I(X;Y) = 0. If they are coupled, I(X;Y) > 0 and the joint entropy is lower than the sum of the marginal entropies.
For more than two sites, it is generally not sufficient to subtract a simple sum of pairwise mutual information terms, because higher-order dependencies may exist. The safe general expression is:

$$H(X_1, X_2, \ldots, X_n) \leq \Sigma_i\ H(X_i)$$

The difference between the sum of marginal entropies and the joint entropy is the total correlation, also called multi-information:

$$TC(X_1,\ldots,X_n) = \Sigma_i\ H(X_i) - H(X_1,\ldots,X_n)$$

$$H(X_1,\ldots,X_n) = \Sigma_i\ H(X_i) - TC(X_1,\ldots,X_n)$$

Total correlation is non-negative and captures the aggregate dependence among sites, including higher-order dependencies that are not reducible to pairwise mutual information.

S2.5. Scope and data requirements for estimating achieved entropy

The examples above show that achieved entropy can be smaller than state-space capacity when the unmodified state is dominant or when specific modified states are rarely attained. A precise estimation of achieved in vivo entropy would require condition-resolved occupancy distributions and joint correlations among sites on the same Pol II molecule, and, for information transfer to transcriptional output, coupling between PTM patterns and transcriptional states.

These requirements remain experimentally challenging; therefore, more precise entropy estimation of Pol II, at present, is outside the scope of our work. Instead, the manuscript reports theoretical state-space capacity and a conservative kinetically addressable estimate.

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A practical way to express the impact of coupling is to define an effective number of independently addressable variables. If a group of sites behaves as a coupled module (e.g., CTD phosphorylation patterns constrained by kinase/phosphatase dynamics and repeat-level coupling), then the achieved entropy scales with the number of effectively independent degrees of freedom, not with the raw number of chemically modifiable residues. Thus, this Supplementary Information defines the experimental data required for future conversion of the present capacity framework into condition-specific entropy estimates.

Supplementary Information S3. Sensitivity analyses of Pol II PTM information-capacity estimates

Here, we evaluate how the estimated **maximum information capacity** of Pol II PTM patterns depends on discrete-state modeling choices. The main text reports an **upper-bound capacity** under a residue-level discrete-state model with mutually exclusive states at each site.

For each site i with s_i mutually exclusive states, the maximal capacity is:

$$I_{\max} = \sum_i \log_2 (s_i), \quad N = \prod_i s_i , \quad I_{\max} = \log_2(N).$$

When multiple PTM types are annotated on the same residue, PTMs are treated as mutually exclusive and combined using:

$$s_{\text{composite}} = 1 + \sum_{j=1}^{k} (s_j - 1).$$

These sensitivity analyses quantify how I_{max} changes under alternative plausible assignments of s_i for PTM classes.

S3.1. Alternative state models tested

(A) CTD Proline cis / trans isomerization
Baseline: $s_{\text{Pro}} = 2$ (cis / trans isomerization)
Conservative: $s_{\text{Pro}} = 1$ (unmodified)

(B) CTD Arginine state granularity
Baseline: site-specific Arg state models as shown in Table 2, with different sets of methylation states for each Arg residue, ranging from s=2 to s=6
Intermediate: $s_{\text{Arg}} = 2$ (unmodified / modified)
Conservative: $s_{\text{Arg}} = 1$ (unmodified)
The intermediate Arg/Lys models are included to define possible alternative granularities but are not used as separate rows in the representative S0–S3 scenario table.

(C) CTD Lysine state granularity
Baseline: site-specific Lys7 state models as used in Table 2, with individual Lys7 residues assigned to curated acetylation/methylation state sets ranging from s=4 to s=7 rather than to a universal Lys7 state model.
Intermediate: $s_{\text{Lys}} = 2$ (unmodified/modified)
Conservative: $s_{\text{Lys}} = 1$ (unmodified)

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(D) Threonine phosphorylation

Baseline: threonine sites were modeled according to their curated state sets, including unmodified/phosphorylated or unmodified/O-GlcNAcylated/phosphorylated where applicable.

Conservative Thr model: threonine phosphorylation states were removed from the model; threonine sites lacking other PTM states were assigned $s_i = 1$ and therefore contributed 0 bits.

S3.2. Sensitivity of Rpb2–Rpb12 capacity (based on Supplementary Tables S2–S12)

Using Supplementary Tables S2–S12, and after consolidating multi-PTM residues into composite sites, the maximal information capacity for Rpb2–Rpb12 under the baseline model is:

$$I_{\text{Rpb2–Rpb12}} = 310.89 \text{ bits.}$$

We then recompute $I_{\text{Rpb2–Rpb12}}$ under the alternative Thr granularity state model while leaving all other PTMs unchanged:

Conservative (Thr $= 1$)

$$I_{\text{Rpb2–Rpb12}} = 270.89 \text{ bits.}$$

Therefore, even in the absence of Thr phosphorylation, the Rpb2–Rpb12 subsystem retains a capacity in the hundreds-of-bits range.

S3.3. Sensitivity of Rpb1 core (non-CTD; Supplementary Table S1)

For the non-CTD Rpb1 core (159 sites), the baseline model yields:

$$I_{\text{Rpb1–core}} = 185.87 \text{ bits.}$$

Under alternative non-CTD state models:

Conservative (Thr $s = 1$):

$$I_{\text{Rpb1–core}} \approx 162.87 \text{ bits.}$$

S3.4. Sensitivity of CTD capacity to state definitions

Baseline CTD capacity (main text):

$$I_{\text{CTD}} = 211.22 \text{ bits.}$$

S3.4.1 CTD without isomerization in Pro sites

Replacing Pro cis/trans from $s = 2$ to $s = 1$ (and keeping all other CTD assumptions unchanged):

$$I_{\text{CTD}} \approx 160.22 \text{ bits.}$$

Thus, the explicit 2-state Pro cis/trans model contributes approximately:

$$\Delta I_{\text{CTD}} \approx 51 \text{ bits.}$$

S3.4.2. CTD “maximally conservative” illustrative variant

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If (i) Prolines are considered as unmodified (s=1), (ii) Thr phosphorylation is removed (s=1), (iii) Lys7 PTM states are collapsed to the unmodified state (s=1), and (iv) Arg residues are pooled as s=1 (unmodified), then:

$$I_{\mathrm{CTD}} \approx 110.68 \text{ bits.}$$

S3.5. Scenario table: combined sensitivity for total Pol II capacity

The total Pol II capacity is:

$$I_{\mathrm{Pol\ II}} = I_{\mathrm{CTD}} + I_{\mathrm{Rpb1-core}} + I_{\mathrm{Rpb2-Rpb12}}$$

Assuming $M = 80{,}200$ Pol II molecules per nucleus:

$$I_{\mathrm{cell}} = M \cdot I_{\mathrm{Pol\ II}}$$

Supplementary Information S3-Table S1. Representative sensitivity scenarios for total Pol II maximal information capacity

Scenario	CTD state model	non-CTD state model (Rpb1-core + Rpb2–Rpb12)	$I_{\mathrm{Pol\ II}}$ (bits per molecule)	Bytes per molecule (I÷8)	I_{Cell} (bits per nucleus)	Bytes per nucleus (I÷8)	MB per nucleus (10^6 bytes)
S0 (baseline)	Baseline	Baseline	707.98	88.50	5.68×10^7	7.10×10^6	7.10
S1 (non-CTD conservative)	Baseline	Thr conservative	644.98	80.62	5.17×10^7	6.47×10^6	6.47
S2 (no isomerization)	Pro cis/trans excluded	Baseline	656.98	82.12	5.27×10^7	6.59×10^6	6.59
S3 (all conservative)	Maximally conservative	Thr conservative	544.44	68.05	4.37×10^7	5.46×10^6	5.46

These scenario definitions correspond to the sections of Supplementary Information S3.1–S3.4. $I_{\mathrm{Pol\ II}}$ = I_{CTD} + $I_{\mathrm{Rpb1\text{-}core}}$ + $I_{\mathrm{Rpb2–Rpb12}}$. Cellular totals use I_{cell} = M · $I_{\mathrm{Pol\ II}}$ with M = 80,200. MB is reported in decimal units (1 MB = 10^6 bytes). This table and all calculations in the text can be regenerated using the Reproducibility Module 18 (Supplementary Information S4)

In brief, across plausible alternative state-model definitions, $I_{\mathrm{Pol\ II}}$ remains in the order of hundreds of bits per molecule, and the implied nuclear-level capacity remains multi-megabyte scale under the same independence/uniform assumptions. These sensitivity analyses therefore support the robustness of the qualitative conclusion (high-dimensional PTM pattern capacity), while clarifying the quantitative dependence on discrete-state granularity choices for specific PTM classes (see Supplementary Information S3-Table S1. Representative sensitivity scenarios for total Pol II maximal information capacity).

S3.6. Link to Supplementary Information S2 (capacity vs achieved entropy)

The sensitivity analyses above quantify how the maximal capacity $I_{\mathrm{max}} = \sum_i \log_2(s_i)$ depends on discrete-state granularity choices. Importantly, these results concern state-space capacity under uniform, independent-site assumptions. As discussed in Supplementary Information S2, biological PTM occupancies are typically non-uniform and sites are often coupled, so the achieved entropy H is generally substantially smaller than I_{max} and further reduced by correlations. Accordingly, S3 evaluates robustness of the theoretical capacity estimate to alternative discrete-state definitions. By contrast, S2

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addresses how non-uniform occupancies and site coupling reduce achieved entropy relative to this capacity. Thus, S2 and S3 address complementary sources of uncertainty: biological occupancy/correlation effects and discrete-state model granularity. The numerical occupancy examples are illustrative and are not used to infer in vivo Pol II entropy.

Supplementary Information S4. Reproducibility package for Pol II PTM information-capacity calculations

Overview

This supplement documents two interactive Python (Jupyter) notebooks: Notebook 1 (*notebook_1.ipynb*, Supplementary Data File 1) and Notebook 2 (*notebook_2.ipynb*, Supplementary Data File 2) that together reproduce every numeric value, table and figure reported in the main text and in Supplementary Information S1 and S3. Notebook 1 contains ten Reproducibility Modules (1–10) that retrieve the archival UniProt[37] UniSave canonical sequences for human RNA Pol II subunits 1-12 and build the integrated post-translational modification (PTM) atlas of human RNA polymerase II from public databases and literature-mined entries, respectively saving these two databases as files *uniprot_polII_seqs_{CURRENT_DATE}.tsv* (Supplementary Data File 3) and *ptms_polII_{CURRENT_DATE}.tsv* (Supplementary Data File 4) in the notebook's working directory. Notebook 2 contains eight Reproducibility Modules (11–18) that load the frozen PTM atlas and the version-pinned UniProt sequences and compute the manuscript's information-theoretic results. Modules are intended to run sequentially within each notebook; their numbering matches the order of the calculations in the manuscript and in Supplementary Information S1 and S3.

For exact, network-independent reproducibility, the frozen archival UniProt UniSave sequences *uniprot_polII_seqs_2026_05_22.tsv* and PTM atlas *ptms_polII_2026_05_22.tsv* snapshots are bundled as Supplementary Data File 3 and Supplementary Data File 4. When both files are present in the working directory, every download module in Notebook 1 detects the cache and skips its network call, and Notebook 2 reads the snapshots directly. The Notebooks' audit asserts are pinned to these snapshots and will fail against new ones regenerated from updated upstream databases.

The package regenerates: (1) the integrated PTM atlas and the version-pinned protein sequence reference (Notebook 1, Modules 1–10); (2) a colour-coded map of the 52 heptads and 10-residue tail of the Rpb1 CTD (Table 1; Module 11); (3) the multiplicative CTD state-space size *N_CTD*, the information capacity *I_CTD*, and the residue-class decomposition of *I_CTD* in Table 2 (Module 12); (5) the information capacity of the Rpb1 non-CTD core region (residues 1–1592) in Supplementary Table S1 (Module 13); (6) per-subunit capacities for Rpb2–Rpb12 in Supplementary Tables S2–S12 (Module 14); (7) the per-molecule and per-nucleus totals *I_total* and *I_nucleus* (Module 15); (8) the multi-timescale physiological capacity *I_physio* with its fast, intermediate and slow layers (Module 16); (9) total unique PTM sites and modified residues in the integrated atlas (Module 17); and (10) the four S0–S3 sensitivity scenarios (Supplementary Information S3.1–S3.4 and Supplementary Information S3-Table S1; Module 18).

Notebook 1. PTM and protein sequence data retrieval

Environment check (optional).

Verifies that the execution environment meets the minimum software dependencies for this notebook (Python ≥ 3.10, pandas ≥ 1.3, requests ≥ 2.20, beautifulsoup4 ≥ 4.9, IPython ≥ 7.0) and aborts with a diagnostic message if any requirement is missing or below the required version, preventing execution errors downstream.

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Reproducibility Module 1. Canonical UniProt sequence retrieval.

Loads the 12 human Pol II protein sequences from a saved cache file (named *uniprot_polII_seqs_*.tsv*) if one exists; otherwise fetches them directly from the UniProt database using fixed, pinned version numbers for each protein. The specific versions are locked in to ensure the output is always identical no matter when or where the code is run. The sequences are stored in a simple lookup table (protein ID → sequence string) that is passed to Module 2 for residue-letter lookups and to Module 8, which breaks the main subunit (P24928) into 52 heptads plus 10-residue C-terminal tail. If no cache was found at startup, the freshly downloaded data is saved to a new cache file for next time.

Reproducibility Module 2. UniProt manually curated PTM annotations.

Fetches manually reviewed modification records for each of the twelve Pol II subunits directly from UniProt, keeping only entries that point to a single, specific position in the protein. The modification name is cleaned up by dropping any extra detail after the first semicolon (e.g. "Phosphoserine; by kinase X" becomes just "Phosphoserine"), and the corresponding amino acid letter is looked up from the sequences loaded in Module 1. The output is a tidy table with one row per (Protein, Residue, Modification) combination — the shared format used by all six download modules.

Reproducibility Module 3. UniProt large-scale proteomics PTMs.

Retrieves modification data assembled from large-scale mass-spectrometry experiments via the EBI Proteins API. Each entry reports a short peptide sequence and the position of the modified residue within it; the code uses those two pieces of information to figure out the residue letter and its position in the full protein. Results are emitted in the shared format.

Reproducibility Module 4. dbPTM experimental PTMs.

Downloads all experiment archives from the dbPTM database, decompresses them and reads them into a standard six-column layout. Only rows belonging to the twelve Pol II accessions are kept. The modified residue letter is taken from the central position (index 10) of the 21-amino-acid flanking window surrounding each modification site. All individual per-archive tables are combined and de-duplicated into a single output in the shared format.

Reproducibility Module 5. iPTMnet integrated PTMs.

Loads the iPTMnet integrated scoring file directly as a tab-separated table, keeping only the Substrate, Residue and PTM columns. Rows are filtered to the twelve canonical Pol II accessions plus the alias ‘P24928-1’ that iPTMnet stores separately despite sequence identity to canonical P24928 — the suffix is stripped so these records merge cleanly with the other sources downstream.

Reproducibility Module 6. PhosphoSitePlus PTMs.

Because PhosphoSitePlus does not offer a free data API, this module scrapes the information directly from its web pages. A built-in lookup table maps each Pol II UniProt accession to its internal PhosphoSitePlus identifier. For each subunit, the module first reads the abbreviation glossary from the protein overview page (e.g. "p" → "Phosphorylation"), then works through the site table page and records every human modification site that is not flagged as belonging to an isoform.

Reproducibility Module 7. GlyGen glycosylation and phosphorylation annotations.

Queries the GlyGen protein database using “{*UniProtKB accession number*}-1” identifiers for each Pol II subunit studied, since that is how GlyGen indexes its entries for those UniProtKB canonical protein isoforms. The response is processed in two passes: first collecting glycosylation entries with a meaningful subtype (such as O-GlcNAcylation), then collecting phosphorylation entries. Three-letter amino acid codes returned by the API are converted to single-letter codes, and the “-1” isoform suffix is stripped from the accession before passing results downstream.

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Reproducibility Module 8. CTD heptad position map.

Cuts out the CTD region of the main Pol II subunit (Rpb1, residues 1593–1970) and divides the heptad-bearing stretch into repeating units by splitting on every tyrosine (every CTD heptad starts with a tyrosine residue), producing 52 CTD heptapeptides and a lookup table mapping each residue position to its heptad number and position within that heptad for use in Modules 9 and 10. The remaining 10 residues form the C-terminal tail. Two built-in checks confirm exactly 52 heptads covering exactly 368 residues, catching any accidental changes to the boundary constants. The module also defines two lookup tables used by later modules: one converting single-letter to three-letter amino acid codes, and one translating modification names into descriptive adjectives (e.g. "Phosphorylation" → "phosphorylated") for use in human-readable labels in Module 10.

Reproducibility Module 9. Cross-database alignment and literature-mined additions.

Takes the six harmonised tables from Modules 2–7 and adds a seventh table of modifications not covered by any of those databases, drawn from the literature: proline isomerization at positions 3 and 6 of heptads matching specific sequences (26 + 25 = 51 sites); O-GlcNAcylation at S1829 and S1896; and citrullination at R1810. A built-in check confirms the literature table contains exactly 54 entries. Each source is then collapsed so there is one modification string per (Substrate, Residue) pair, and all seven sources are horizontally joined into a wide audit table (*ptm_comparison*) with one column per data source, so every site reported by any database is visible in one place. The module also counts how many residues carry each unique combination of modifications, for downstream auditing.

Reproducibility Module 10. PTM data curation and integration.

Standardises the modification vocabulary across all sources by collapsing synonyms and inconsistent naming into a single controlled set (e.g. "Ubiquitinylation" and "Ubiquitylation" both become "Ubiquitination"; various phosphoresidue names all become "Phosphorylation"). Two irreversible modification types that do not fit the reversible site model underlying the information-capacity calculation — N-Glycosylation and Caspase cleavage — are removed. Where both a specific O-GlcNAcylation and a generic O-Glycosylation call appear at the same residue, the generic one is absorbed into the specific one to avoid double-counting. The cleaned records are combined and de-duplicated into *all_databases_merge*, the single master table used by all downstream modules. Finally, the master table is written to disk as ptms_polII_{CURRENT DATE}.tsv; on future runs all download and curation steps detect this file and skip all network access entirely, preserving reproducibility without re-querying any database.

Notebook 2. Information-theoretic calculations

Environment check and Analysis setup (required).

The environment check confirms that the required software versions are present and halts with a clear message if anything is missing. The analysis setup loads the most recent sequence cache and PTM master table from disk, reconstructs the 52-heptad position map for Rpb1 using the same boundary constants as Module 8 in Notebook 1, creates the *ctd_subclass* dictionary, and builds the amino acid and modification name lookup tables in the same way as Module 8 in Notebook 1. It then splits *all_databases_merge* into three non-overlapping pandas dataframes used by all downstream modules: *ctd_df* for Rpb1 CTD (positions 1593–1970), *core_df* for Rpb1 non-CTD core (positions 1–1592) and *r212_df* for the eleven non-Rpb1 subunits. It also defines the shared helper function *_I* that computes information capacity in bits as a sum of log2(*number of modification states + 1*), where +1 accounts for the baseline unmodified state, over all modified residues. Running this cell first is mandatory, as everything else in Notebook 2 depends on it.

Reproducibility Module 11. Colour-coded CTD heptad map (Table 1).

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Produces Table 1 from the main text: an HTML rendering of all 52 CTD heptapeptides plus the 10-residue C-terminal tail, with each residue coloured according to which PTMs have been reported on it. The colour palette is built automatically from the combinations observed in the data, and a legend is included. Two non-standard CTD heptads carrying extra residues are marked with an asterisk.

Reproducibility Module 12. CTD combinatorial state space, information capacity, and Table 2 of the manuscript.

Calculates how many distinct modifications states the CTD can theoretically encode by multiplying together the number of possible states at each site (including the unmodified state), then takes the $\log_2$ of that product to express the result in bits. Both numbers are checked against the values reported in the manuscript ($\log_{10}$ of *N_CTD* = 63.58; *N_CTD* = 211.22 bits) and will raise an error if recalculated from a different data snapshot. The module then produces Table 2, breaking the total capacity down by residue class and modification type, with the heptad region's residues listed first in biological order and the C-terminal tail appended at the end. A final check confirms the per-row bit contributions sum exactly to *I_CTD*.

Reproducibility Module 13. Information capacity of the Rpb1 non-CTD core (Supplementary Table S1).

Applies the same capacity calculation to the non-CTD (core) portion of Rpb1 (residues 1–1592) and renders Supplementary Table S1, with one row per modified residue sorted by position. Columns show the residue, PTM type(s), UniProt accession, number of possible states, and bits per site. Summary counts of unique PTM types, modified residues, and (residue, PTM type) pairs are reported, with a check that the total calculated information capacity matches the manuscript value and the per-site bits sum to the same correct total.

Reproducibility Module 14. Information capacity encoded by PTMs in subunits Rpb2–Rpb12 (Supplementary Tables S2–S12).

Repeats the same procedure for the eleven non-Rpb1 subunits and produces one supplementary table per subunit in canonical order (Rpb2 through Rpb12). Each table lists every modified residue, its PTM combination, its UniProt accession, its per-site number of possible states, and its per-site bit contribution. Per-subunit capacities are printed and audited against their respective values reported in the manuscript, and final checks confirm that the computed total information capacity for Rpb1-Rpb12 matches both the manuscript value and the sum of the per-site bit contribution column across all tables.

Reproducibility Module 15. Per-molecule and per-nucleus information capacity.

Adds the three compartment capacities (CTD + Rpb1 core + Rpb2–12) to get the total per-molecule *I_total*, then scales up by *N_POLII* = 80,200 — the estimated number of Pol II complexes in a mammalian nucleus (BioNumbers ID 112321; Zhao et al., 2014)[63] — to get an upper-bound nuclear capacity. Results are reported in bits, bytes, and megabytes. A built-in check guards computed per-molecule and per-nucleus values against drifting from the manuscript value.

Reproducibility Module 16. Multi-timescale PTM biochemical architecture.

Divides the per-molecule capacity into three layers defined by their characteristic regulatory timescales: a fast layer of 30 phosphorylation sites cycling on a ∼5 s–15 min timescale with 5 effective states each; an intermediate layer of 32 proline isomerization and O-GlcNAcylation sites cycling over ∼1–30 min with 2 states each; and a slow layer of 10 ubiquitination-related sites turning over on ∼15 min–∼24 h with 2.5 effective states each. The three-layer capacities are summed into a physiological per-molecule total, also reported at nuclear scale and compared to the theoretical upper bound. Checks confirm the physiological information capacity per molecule and per nucleus match manuscript values.

Reproducibility Module 17. Total unique PTM sites and modified residues (Supplementary Information S1).

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Reports the total number of unique modified residues, counted as distinct (Accession, Residue) pairs, and the total number of unique (Accession, Residue, PTM) triplets across the entire integrated PTM atlas. These two scalars are then checked against the totals quoted in Supplementary Information S1.

Reproducibility Module 18. Sensitivity analyses (Supplementary Information S3 and Supplementary Information S3-Table S1).

Tests how sensitive the main results are to modelling assumptions by recalculating total Pol II information capacity under four scenarios: the manuscript baseline (S0); a conservative version that removes threonine phosphorylation outside the CTD (S1); a version that keeps the non-CTD model but removes proline isomerization from the CTD (S2); and a maximally conservative version that removes proline isomerization, threonine phosphorylation everywhere, and all lysine-7 and arginine modifications in the CTD (S3). Per-molecule and per-nucleus information capacities are reported for each scenario and assembled into Supplementary Table S3. Checks pin each scenario total to its manuscript value.

Supplementary Tables S1–S12, note

PTM sites were compiled from UniProt, dbPTM, iPTMnet, PhosphoSitePlus and GlyGen. For each residue, we assign a discrete state count s_i, with s=2 for every PTM type (phosphorylation s=2, acetylation s=2, etc.). When multiple PTM types map to the same residue, PTMs are treated as mutually exclusive and combined using: $s_{\text{composite}} = 1 + \sum_{j=1}^{k}(s_j - 1)$. Bits per site are $log_2(s_i)$.

Supplementary Table S1 POLR2A (RPB1 core, non-CTD)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $log_2(s_i)$
M1	Acetylation	P24928	2	1.00
S8	Phosphorylation	P24928	2	1.00
S27	Phosphorylation	P24928	2	1.00
K32	SUMOylation / Ubiquitination	P24928	3	1.58
S35	Phosphorylation	P24928	2	1.00
T37	Phosphorylation	P24928	2	1.00
K42	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
T46	Phosphorylation	P24928	2	1.00
T47	Phosphorylation	P24928	2	1.00
K53	Ubiquitination	P24928	2	1.00
K92	Ubiquitination	P24928	2	1.00
K116	Ubiquitination	P24928	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S121	Phosphorylation	P24928	2	1.00
K125	Ubiquitination	P24928	2	1.00
K127	Ubiquitination	P24928	2	1.00
K132	Ubiquitination	P24928	2	1.00
K134	Acetylation	P24928	2	1.00
K139	Acetylation	P24928	2	1.00
Y145	Phosphorylation	P24928	2	1.00
K149	SUMOylation / Ubiquitination	P24928	3	1.58
K151	SUMOylation / Ubiquitination	P24928	3	1.58
K163	SUMOylation / Ubiquitination	P24928	3	1.58
T176	Phosphorylation	P24928	2	1.00
K177	Acetylation / Ubiquitination	P24928	3	1.58
K179	Ubiquitination	P24928	2	1.00
Y199	Phosphorylation	P24928	2	1.00
K203	Ubiquitination	P24928	2	1.00
S209	Phosphorylation	P24928	2	1.00
K212	Ubiquitination	P24928	2	1.00
K213	SUMOylation / Ubiquitination	P24928	3	1.58
S217	Phosphorylation	P24928	2	1.00
K226	Ubiquitination	P24928	2	1.00
C233	Glutathionylation / Methylation	P24928	3	1.58
S269	Phosphorylation	P24928	2	1.00
K279	SUMOylation / Ubiquitination	P24928	3	1.58
K285	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
K308	SUMOylation	P24928	2	1.00
K331	Ubiquitination	P24928	2	1.00
K337	Ubiquitination	P24928	2	1.00
K340	Ubiquitination	P24928	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K357	Acetylation / Ubiquitination	P24928	3	1.58
T368	Phosphorylation	P24928	2	1.00
S374	Phosphorylation	P24928	2	1.00
K417	Ubiquitination	P24928	2	1.00
R426	Methylation	P24928	2	1.00
K434	Ubiquitination	P24928	2	1.00
K445	SUMOylation / Ubiquitination	P24928	3	1.58
C451	Glutathionylation	P24928	2	1.00
K466	SUMOylation / Ubiquitination	P24928	3	1.58
T481	Phosphorylation	P24928	2	1.00
M520	Sulfoxidation	P24928	2	1.00
S530	Phosphorylation	P24928	2	1.00
T543	Phosphorylation	P24928	2	1.00
T549	Phosphorylation	P24928	2	1.00
K581	SUMOylation / Ubiquitination	P24928	3	1.58
T587	Phosphorylation	P24928	2	1.00
S607	Phosphorylation	P24928	2	1.00
T608	Phosphorylation	P24928	2	1.00
Y618	Phosphorylation	P24928	2	1.00
K619	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
K627	SUMOylation / Ubiquitination	P24928	3	1.58
K642	Acetylation / Ubiquitination	P24928	3	1.58
K643	Ubiquitination	P24928	2	1.00
K697	Ubiquitination	P24928	2	1.00
T698	Phosphorylation	P24928	2	1.00
Y699	Phosphorylation	P24928	2	1.00
K707	SUMOylation / Ubiquitination	P24928	3	1.58
K708	Methylation / Ubiquitination	P24928	3	1.58

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K710	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
K719	Ubiquitination	P24928	2	1.00
T732	Phosphorylation	P24928	2	1.00
T736	Phosphorylation	P24928	2	1.00
K751	Ubiquitination	P24928	2	1.00
K758	Ubiquitination	P24928	2	1.00
S759	Phosphorylation	P24928	2	1.00
S761	Phosphorylation	P24928	2	1.00
Y763	Phosphorylation	P24928	2	1.00
K767	Methylation / SUMOylation / Ubiquitination	P24928	4	2.00
S768	Phosphorylation	P24928	2	1.00
S772	Phosphorylation	P24928	2	1.00
K775	Ubiquitination	P24928	2	1.00
S777	Phosphorylation	P24928	2	1.00
K778	SUMOylation / Ubiquitination	P24928	3	1.58
K796	SUMOylation / Ubiquitination	P24928	3	1.58
K803	Acetylation / Methylation / SUMOylation / Ubiquitination	P24928	5	2.32
K812	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
K853	SUMOylation / Ubiquitination	P24928	3	1.58
T857	Phosphorylation	P24928	2	1.00
Y859	Phosphorylation	P24928	2	1.00
K866	SUMOylation / Ubiquitination	P24928	3	1.58
S867	Phosphorylation	P24928	2	1.00
S870	Phosphorylation	P24928	2	1.00
K874	Ubiquitination	P24928	2	1.00
Y875	Phosphorylation	P24928	2	1.00
T878	Phosphorylation	P24928	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S882	Phosphorylation	P24928	2	1.00
K910	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
K914	Ubiquitination	P24928	2	1.00
K918	Ubiquitination	P24928	2	1.00
K919	Ubiquitination	P24928	2	1.00
K940	Acetylation / SUMOylation / Ubiquitination	P24928	4	2.00
T972	Phosphorylation	P24928	2	1.00
S975	Phosphorylation	P24928	2	1.00
K976	SUMOylation / Ubiquitination	P24928	3	1.58
K992	Ubiquitination	P24928	2	1.00
K1008	Acetylation / Ubiquitination	P24928	3	1.58
K1014	Acetylation / Ubiquitination	P24928	3	1.58
K1018	Ubiquitination	P24928	2	1.00
K1019	Ubiquitination	P24928	2	1.00
S1030	Phosphorylation	P24928	2	1.00
C1050	Methylation	P24928	2	1.00
Y1109	Phosphorylation	P24928	2	1.00
K1125	SUMOylation / Ubiquitination	P24928	3	1.58
K1132	Ubiquitination	P24928	2	1.00
K1133	Ubiquitination	P24928	2	1.00
K1135	SUMOylation / Ubiquitination	P24928	3	1.58
S1138	Phosphorylation	P24928	2	1.00
S1147	Phosphorylation	P24928	2	1.00
K1155	Ubiquitination	P24928	2	1.00
C1159	Methylation	P24928	2	1.00
K1219	Ubiquitination	P24928	2	1.00
K1225	SUMOylation / Ubiquitination	P24928	3	1.58
K1234	Ubiquitination	P24928	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K1254	SUMOylation / Ubiquitination	P24928	3	1.58
K1268	SUMOylation / Ubiquitination	P24928	3	1.58
K1278	Ubiquitination	P24928	2	1.00
C1287	Glutathionylation	P24928	2	1.00
K1306	Ubiquitination	P24928	2	1.00
Y1308	Phosphorylation	P24928	2	1.00
K1317	Ubiquitination	P24928	2	1.00
K1318	Ubiquitination	P24928	2	1.00
K1319	Ubiquitination	P24928	2	1.00
K1329	Ubiquitination	P24928	2	1.00
S1348	Phosphorylation	P24928	2	1.00
K1350	Acetylation / Ubiquitination	P24928	3	1.58
Y1383	Phosphorylation	P24928	2	1.00
S1387	Phosphorylation	P24928	2	1.00
S1391	Phosphorylation	P24928	2	1.00
Y1392	Phosphorylation	P24928	2	1.00
Y1395	Phosphorylation	P24928	2	1.00
T1406	Phosphorylation	P24928	2	1.00
T1415	Phosphorylation	P24928	2	1.00
T1424	Phosphorylation	P24928	2	1.00
K1429	Ubiquitination	P24928	2	1.00
K1479	Ubiquitination	P24928	2	1.00
K1481	Ubiquitination	P24928	2	1.00
S1505	Phosphorylation	P24928	2	1.00
S1508	Phosphorylation	P24928	2	1.00
S1514	Phosphorylation	P24928	2	1.00
T1518	Phosphorylation	P24928	2	1.00
T1525	Phosphorylation	P24928	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S1532	Phosphorylation	P24928	2	1.00
S1534	O-GlcNAcylation / Phosphorylation	P24928	3	1.58
T1540	Phosphorylation	P24928	2	1.00
T1568	Phosphorylation	P24928	2	1.00
S1571	Phosphorylation	P24928	2	1.00
S1574	Phosphorylation	P24928	2	1.00
S1590	Phosphorylation	P24928	2	1.00
S1592	Phosphorylation	P24928	2	1.00

Supplementary Table S2 POLR2B (RPB2 / Rpb2)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K37	Ubiquitination	P30876	2	1.00
S75	Phosphorylation	P30876	2	1.00
K87	SUMOylation / Ubiquitination	P30876	3	1.58
Y92	Phosphorylation	P30876	2	1.00
S94	Phosphorylation	P30876	2	1.00
K95	Acetylation / Ubiquitination	P30876	3	1.58
T97	Phosphorylation	P30876	2	1.00
S106	Phosphorylation	P30876	2	1.00
T119	Phosphorylation	P30876	2	1.00
Y120	Phosphorylation	P30876	2	1.00
S121	Phosphorylation	P30876	2	1.00
Y125	Phosphorylation	P30876	2	1.00
K130	Ubiquitination	P30876	2	1.00
K134	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
K146	Ubiquitination	P30876	2	1.00
T147	Phosphorylation	P30876	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K151	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
C172	Methylation	P30876	2	1.00
K192	Ubiquitination	P30876	2	1.00
K199	SUMOylation / Ubiquitination	P30876	3	1.58
T202	Phosphorylation	P30876	2	1.00
T204	Phosphorylation	P30876	2	1.00
Y206	Phosphorylation	P30876	2	1.00
K210	Ubiquitination	P30876	2	1.00
K211	Ubiquitination	P30876	2	1.00
S213	Phosphorylation	P30876	2	1.00
K214	Ubiquitination	P30876	2	1.00
Y215	Phosphorylation	P30876	2	1.00
Y217	Phosphorylation	P30876	2	1.00
T218	Phosphorylation	P30876	2	1.00
C221	Methylation	P30876	2	1.00
S228	Phosphorylation	P30876	2	1.00
S233	Phosphorylation	P30876	2	1.00
T234	Phosphorylation	P30876	2	1.00
K248	Ubiquitination	P30876	2	1.00
K249	Ubiquitination	P30876	2	1.00
K264	SUMOylation / Ubiquitination	P30876	3	1.58
K327	Ubiquitination	P30876	2	1.00
K332	Ubiquitination	P30876	2	1.00
K337	Ubiquitination	P30876	2	1.00
K340	Ubiquitination	P30876	2	1.00
K345	Ubiquitination	P30876	2	1.00
S354	Phosphorylation	P30876	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
T359	Phosphorylation	P30876	2	1.00
K360	Acetylation / Ubiquitination	P30876	3	1.58
K361	Ubiquitination	P30876	2	1.00
K391	Ubiquitination	P30876	2	1.00
K409	Ubiquitination	P30876	2	1.00
K413	Ubiquitination	P30876	2	1.00
K421	Ubiquitination	P30876	2	1.00
K427	Ubiquitination	P30876	2	1.00
K436	Ubiquitination	P30876	2	1.00
T437	Phosphorylation	P30876	2	1.00
S441	Phosphorylation	P30876	2	1.00
K445	Methylation / Ubiquitination	P30876	3	1.58
K457	Ubiquitination	P30876	2	1.00
K458	Acetylation / Ubiquitination	P30876	3	1.58
S467	Phosphorylation	P30876	2	1.00
S487	Phosphorylation	P30876	2	1.00
K494	Ubiquitination	P30876	2	1.00
K497	Ubiquitination	P30876	2	1.00
K578	Acetylation / Ubiquitination	P30876	3	1.58
T586	Phosphorylation	P30876	2	1.00
C622	Methylation	P30876	2	1.00
K630	Ubiquitination	P30876	2	1.00
K632	Ubiquitination	P30876	2	1.00
K637	Ubiquitination	P30876	2	1.00
K644	SUMOylation / Ubiquitination	P30876	3	1.58
Y689	Phosphorylation	P30876	2	1.00
C696	Methylation	P30876	2	1.00
Y724	Phosphorylation	P30876	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K730	Methylation / Ubiquitination	P30876	3	1.58
Y736	Phosphorylation	P30876	2	1.00
T746	Phosphorylation	P30876	2	1.00
Y752	Phosphorylation	P30876	2	1.00
Y753	Phosphorylation	P30876	2	1.00
K756	Ubiquitination	P30876	2	1.00
T760	Phosphorylation	P30876	2	1.00
T761	Phosphorylation	P30876	2	1.00
S763	Phosphorylation	P30876	2	1.00
R803	Methylation	P30876	2	1.00
K815	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
K820	Acetylation	P30876	2	1.00
K821	Ubiquitination	P30876	2	1.00
K831	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
Y845	Phosphorylation	P30876	2	1.00
K847	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
K869	SUMOylation / Ubiquitination	P30876	3	1.58
T870	Phosphorylation	P30876	2	1.00
T872	Phosphorylation	P30876	2	1.00
K889	Ubiquitination	P30876	2	1.00
K917	Ubiquitination	P30876	2	1.00
C919	Methylation	P30876	2	1.00
K920	Ubiquitination	P30876	2	1.00
S925	Phosphorylation	P30876	2	1.00
K934	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
S937	Phosphorylation	P30876	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K942	SUMOylation / Ubiquitination	P30876	3	1.58
C945	Methylation	P30876	2	1.00
K988	Ubiquitination	P30876	2	1.00
K993	Acetylation / SUMOylation / Ubiquitination	P30876	4	2.00
T1000	Phosphorylation	P30876	2	1.00
K1010	SUMOylation / Ubiquitination	P30876	3	1.58
S1012	Phosphorylation	P30876	2	1.00
S1016	Phosphorylation	P30876	2	1.00
Y1018	Phosphorylation	P30876	2	1.00
Y1020	Phosphorylation	P30876	2	1.00
R1023	Methylation	P30876	2	1.00
Y1029	Phosphorylation	P30876	2	1.00
T1033	Phosphorylation	P30876	2	1.00
K1036	SUMOylation / Ubiquitination	P30876	3	1.58
T1046	O-GlcNAcylation	P30876	2	1.00
K1052	Methylation / Ubiquitination	P30876	3	1.58
K1058	Acetylation / Ubiquitination	P30876	3	1.58
R1064	Methylation	P30876	2	1.00
S1079	Phosphorylation	P30876	2	1.00
C1093	Methylation	P30876	2	1.00
C1119	Methylation	P30876	2	1.00
K1143	Ubiquitination	P30876	2	1.00
T1144	Phosphorylation	P30876	2	1.00
S1147	Phosphorylation	P30876	2	1.00
C1155	Methylation	P30876	2	1.00
K1156	Ubiquitination	P30876	2	1.00
S1166	Phosphorylation	P30876	2	1.00

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Supplementary Table S3 POLR2C (RPB3 / Rpb3)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
Y3	Phosphorylation	P19387	2	1.00
T8	Phosphorylation	P19387	2	1.00
T12	Phosphorylation	P19387	2	1.00
T15	Phosphorylation	P19387	2	1.00
K20	SUMOylation / Ubiquitination	P19387	3	1.58
T26	Phosphorylation	P19387	2	1.00
S33	Phosphorylation	P19387	2	1.00
K81	SUMOylation / Ubiquitination	P19387	3	1.58
S122	Phosphorylation	P19387	2	1.00
S124	Phosphorylation	P19387	2	1.00
Y142	Phosphorylation	P19387	2	1.00
K152	Ubiquitination	P19387	2	1.00
K171	SUMOylation	P19387	2	1.00
K175	SUMOylation / Ubiquitination	P19387	3	1.58
K199	SUMOylation / Ubiquitination	P19387	3	1.58
K205	SUMOylation / Ubiquitination	P19387	3	1.58
S206	Phosphorylation	P19387	2	1.00
Y208	Phosphorylation	P19387	2	1.00
S209	Phosphorylation	P19387	2	1.00
S216	Phosphorylation	P19387	2	1.00
Y220	Phosphorylation	P19387	2	1.00
K225	Acetylation / SUMOylation / Ubiquitination	P19387	4	2.00
S235	Phosphorylation	P19387	2	1.00
K253	Ubiquitination	P19387	2	1.00
K254	Ubiquitination	P19387	2	1.00
K255	SUMOylation / Ubiquitination	P19387	3	1.58

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S257	Phosphorylation	P19387	2	1.00
T261	Phosphorylation	P19387	2	1.00
S264	Phosphorylation	P19387	2	1.00

Supplementary Table S4 POLR2D (RPB4 / Rpb4)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S18	Phosphorylation	O15514	2	1.00
T28	Phosphorylation	O15514	2	1.00
S35	Phosphorylation	O15514	2	1.00
K45	Ubiquitination	O15514	2	1.00
Y67	Phosphorylation	O15514	2	1.00
S72	Phosphorylation	O15514	2	1.00
K91	Ubiquitination	O15514	2	1.00
K94	Acetylation	O15514	2	1.00
K112	Ubiquitination	O15514	2	1.00
K137	Ubiquitination	O15514	2	1.00

Supplementary Table S5 POLR2E (RPABC1 / Rpb5)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
M1	Acetylation	P19388	2	1.00
T7	Phosphorylation	P19388	2	1.00
Y8	Phosphorylation	P19388	2	1.00
R9	Methylation	P19388	2	1.00
K15	Ubiquitination	P19388	2	1.00
T29	Phosphorylation	P19388	2	1.00
T56	Phosphorylation	P19388	2	1.00
T59	Phosphorylation	P19388	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K81	SUMOylation / Ubiquitination	P19388	3	1.58
K85	Ubiquitination	P19388	2	1.00
K88	Acetylation / Ubiquitination	P19388	3	1.58
Y90	Phosphorylation	P19388	2	1.00
M110	Sulfoxidation	P19388	2	1.00
T111	Phosphorylation	P19388	2	1.00
S113	Phosphorylation	P19388	2	1.00
K115	SUMOylation / Ubiquitination	P19388	3	1.58
K153	SUMOylation / Ubiquitination	P19388	3	1.58
T157	Phosphorylation	P19388	2	1.00
K186	Ubiquitination	P19388	2	1.00
K192	Acetylation / SUMOylation / Ubiquitination	P19388	4	2.00
S197	Phosphorylation	P19388	2	1.00
T199	Phosphorylation	P19388	2	1.00

Supplementary Table S6 POLR2F (RPABC2 / Rpb6)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
S2	Acetylation / Phosphorylation	P61218	3	1.58
S41	Phosphorylation	P61218	2	1.00
T54	Phosphorylation	P61218	2	1.00
K59	Ubiquitination	P61218	2	1.00
K101	Ubiquitination	P61218	2	1.00

Supplementary Table S7 POLR2G (RPB7 / Rpb7)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
Y3	Phosphorylation	P62487	2	1.00
S6	Phosphorylation	P62487	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
Y17	Phosphorylation	P62487	2	1.00
K27	Ubiquitination	P62487	2	1.00
K29	Acetylation / SUMOylation / Ubiquitination	P62487	4	2.00
C38	Methylation	P62487	2	1.00
K71	Ubiquitination	P62487	2	1.00
K73	SUMOylation / Ubiquitination	P62487	3	1.58
K81	Acetylation / SUMOylation / Ubiquitination	P62487	4	2.00
K94	Ubiquitination	P62487	2	1.00
C106	Methylation	P62487	2	1.00
K129	SUMOylation / Ubiquitination	P62487	3	1.58
M131	Sulfoxidation	P62487	2	1.00
K146	Ubiquitination	P62487	2	1.00
K154	Ubiquitination	P62487	2	1.00

Supplementary Table S8 POLR2H (RPABC3 / Rpb8)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
A2	Acetylation	P52434	2	1.00
K13	SUMOylation / Ubiquitination	P52434	3	1.58
K20	Acetylation / SUMOylation / Ubiquitination	P52434	4	2.00
K21	Ubiquitination	P52434	2	1.00
C30	Methylation	P52434	2	1.00
K36	Acetylation	P52434	2	1.00
K59	Ubiquitination	P52434	2	1.00
Y65	Phosphorylation	P52434	2	1.00
Y75	Phosphorylation	P52434	2	1.00
T78	Phosphorylation	P52434	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
K82	Ubiquitination	P52434	2	1.00
K83	Ubiquitination	P52434	2	1.00
Y90	Phosphorylation	P52434	2	1.00
M92	Sulfoxidation	P52434	2	1.00
K95	Ubiquitination	P52434	2	1.00
T104	Phosphorylation	P52434	2	1.00
S105	Phosphorylation	P52434	2	1.00
K110	Ubiquitination	P52434	2	1.00
K111	Ubiquitination	P52434	2	1.00
K118	Ubiquitination	P52434	2	1.00
Y118	Phosphorylation	P52434	2	1.00
K119	Ubiquitination	P52434	2	1.00
Y142	Phosphorylation	P52434	2	1.00
K146	Ubiquitination	P52434	2	1.00
K147	Ubiquitination	P52434	2	1.00

Supplementary Table S9 POLR2I (RPB9 / Rpb9)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
M1	Acetylation	P36954	2	1.00
Y7	Phosphorylation	P36954	2	1.00
K27	Ubiquitination	P36954	2	1.00
Y37	Phosphorylation	P36954	2	1.00
S51	Phosphorylation	P36954	2	1.00
Y54	Phosphorylation	P36954	2	1.00
T59	Phosphorylation	P36954	2	1.00
T66	Phosphorylation	P36954	2	1.00
S73	Phosphorylation	P36954	2	1.00
T77	Phosphorylation	P36954	2	1.00

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
T81	Phosphorylation	P36954	2	1.00
S101	Phosphorylation	P36954	2	1.00
Y111	Phosphorylation	P36954	2	1.00
Y112	Phosphorylation	P36954	2	1.00
T115	Phosphorylation	P36954	2	1.00

Supplementary Table S10 POLR2L (RPABC5 / Rpb10)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
C10	Methylation	P62875	2	1.00
K12	Ubiquitination	P62875	2	1.00
K17	Ubiquitination	P62875	2	1.00
K41	Ubiquitination	P62875	2	1.00
M48	Methylation	P62875	2	1.00
K58	Ubiquitination	P62875	2	1.00
Y62	Phosphorylation	P62875	2	1.00
K67	Ubiquitination	P62875	2	1.00

Supplementary Table S11 POLR2J (RPB11-a / Rpb11)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
M1	Acetylation	P52435	2	1.00
S9	Phosphorylation	P52435	2	1.00
K17	Ubiquitination	P52435	2	1.00
K22	Ubiquitination	P52435	2	1.00
K23	Ubiquitination	P52435	2	1.00
K25	Ubiquitination	P52435	2	1.00
K26	SUMOylation / Ubiquitination	P52435	3	1.58
K32	Ubiquitination	P52435	2	1.00
K37	Acetylation / Ubiquitination	P52435	3	1.58

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Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
T41	Phosphorylation	P52435	2	1.00
K46	Ubiquitination	P52435	2	1.00
K47	SUMOylation / Ubiquitination	P52435	3	1.58
K51	Ubiquitination	P52435	2	1.00
K52	Ubiquitination	P52435	2	1.00
K61	Ubiquitination	P52435	2	1.00
Y61	Phosphorylation	P52435	2	1.00
K62	Acetylation / SUMOylation / Ubiquitination	P52435	4	2.00
K70	Acetylation / Ubiquitination	P52435	3	1.58
S99	Phosphorylation	P52435	2	1.00
K109	Ubiquitination	P52435	2	1.00
K110	Ubiquitination	P52435	2	1.00
K112	SUMOylation	P52435	2	1.00

Supplementary Table S12 POLR2K (RPABC4 / Rpb12)

Residue	PTM Type(s)	UniProt accession	States (s_i)	Bits per site $\log_2(s_i)$
M1	Acetylation	P53803	2	1.00
K5	Ubiquitination	P53803	2	1.00
C22	Methylation	P53803	2	1.00